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Gotschall et al.

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(54) **SYSTEM FOR TRACKING AND MANAGEMENT OF EMS RESPONSES**

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(73) Assignee: **ZOLL MEDICAL CORPORATION**, Chelmsford, MA (US)

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Related U.S. Application Data

(63) Continuation of application No. 17/152,133, filed on Jan. 19, 2021, now Pat. No. 11,635,300, which is a continuation of application No. 16/786,203, filed on Feb. 10, 2020, now Pat. No. 10,942,040, which is a continuation of application No. 13/467,859, filed on May 9, 2012, now Pat. No. 10,598,508.

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(51) **Int. Cl.**
G01C 21/36 (2006.01)

(52) **U.S. Cl.**
CPC ***G01C 21/3664*** (2013.01)

(58) **Field of Classification Search**
CPC G06F 16/29
See application file for complete search history.

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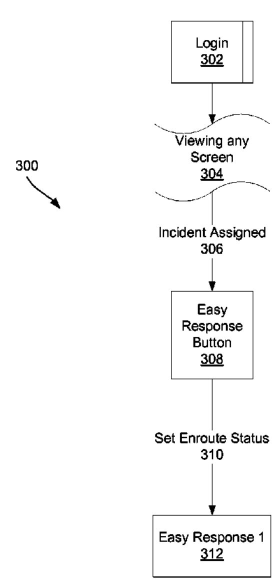
Primary Examiner — Di Xiao
(74) *Attorney, Agent, or Firm* — Finch & Maloney PLLC

(57) **ABSTRACT**

A system for tracking and management of emergency medical services (EMS) responses includes one or more servers configured to assign two or more EMS responses of a plurality of EMS responses corresponding to a plurality of incidences to an EMS vehicle, wherein a first EMS response of the two or more EMS responses is an emergency patient transport and another EMS response of the two or more EMS responses is a pre-scheduled patient transport, provide a user interface comprising a map comprising location information for the two or more EMS responses, receive location information updates for the EMS vehicle, update the map and a status of the EMS vehicle at the user interface based on the location information updates, and store a status history for the EMS vehicle.

21 Claims, 25 Drawing Sheets



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709/206
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705/2
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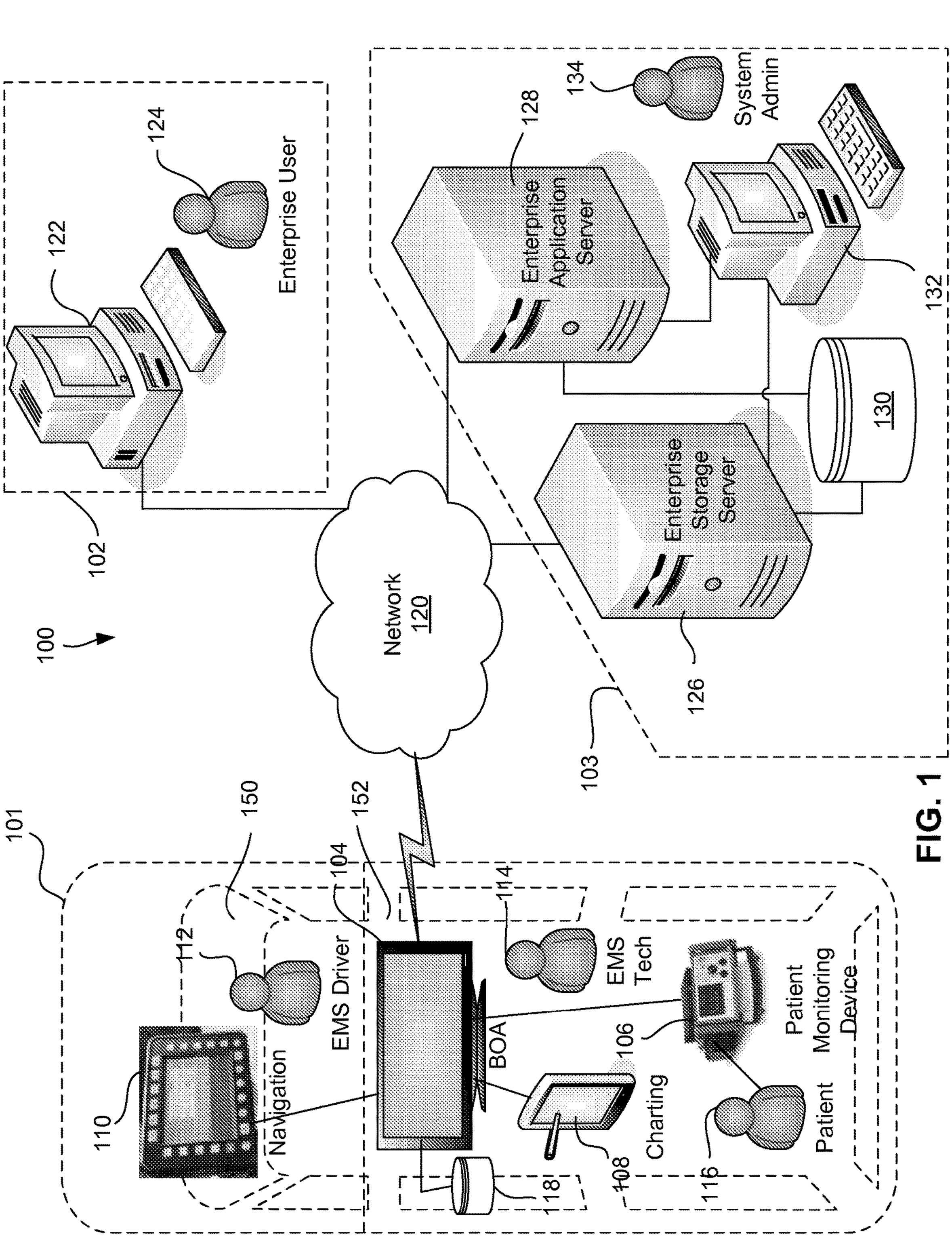
100
101
102
103
Enterprise User
122
124
Network
120
Enterprise Application Server
128
Enterprise Storage Server
126
130
System Admin
132
134
150
152
104
114
112
110
Navigation
EMS Driver
BOA
EMS Tech
106
Patient Monitoring Device
Charting
108
118
116
Patient

FIG. 1

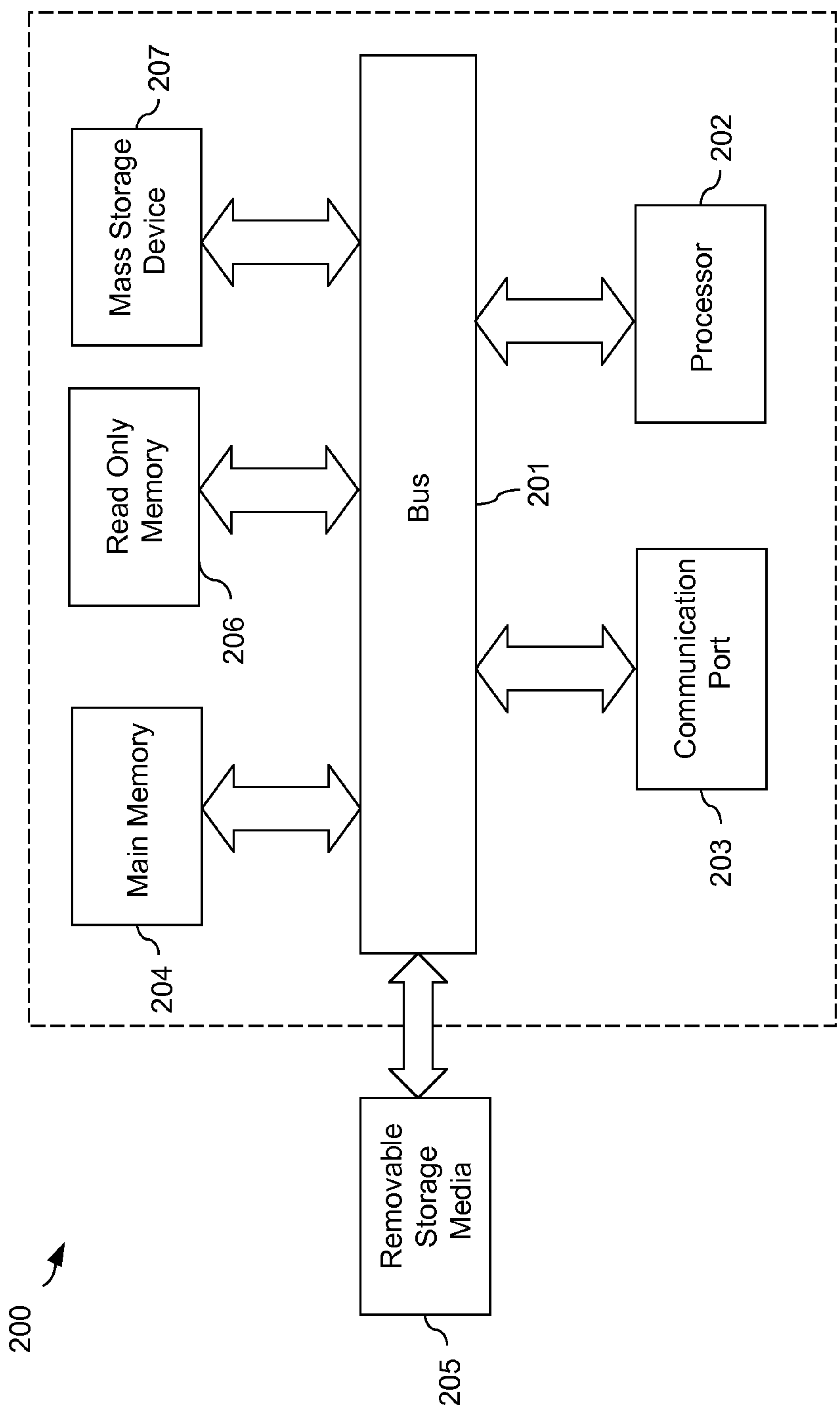
200
Mass Storage Device
207
Read Only Memory
206
Main Memory
204
Bus
201
Processor
202
Communication Port
203
Removable Storage Media
205

FIG. 2

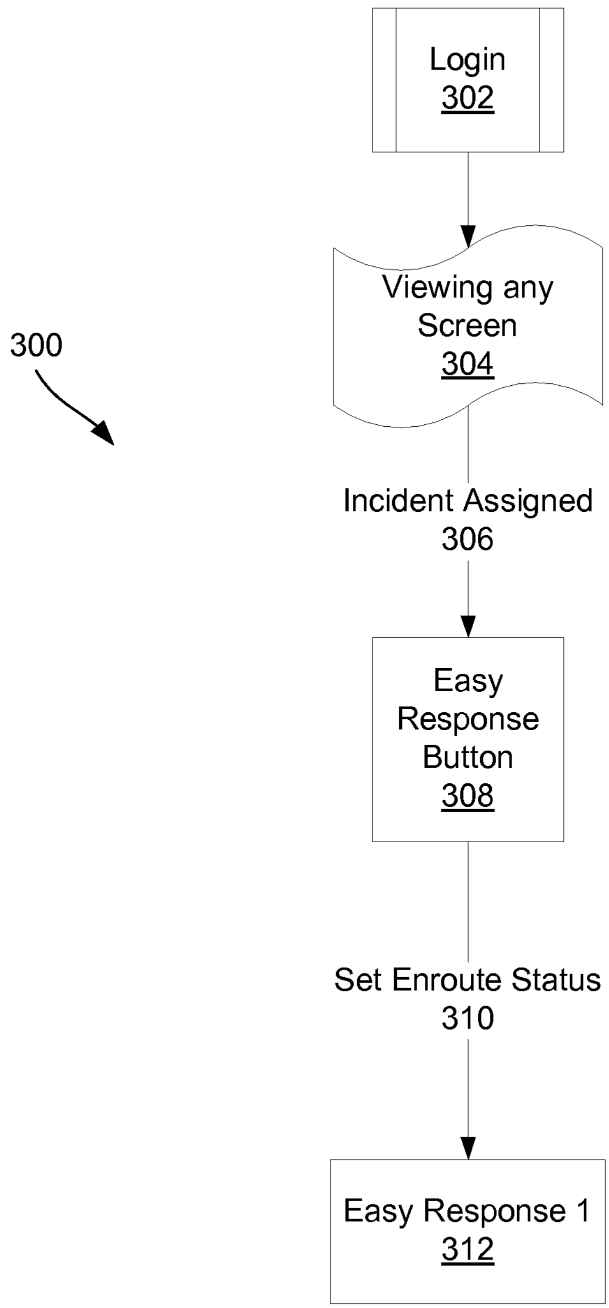
Login
302
Viewing any Screen
304
300
Incident Assigned
306
Easy Response Button
308
Set Enroute Status
310
Easy Response 1
312

FIG. 3

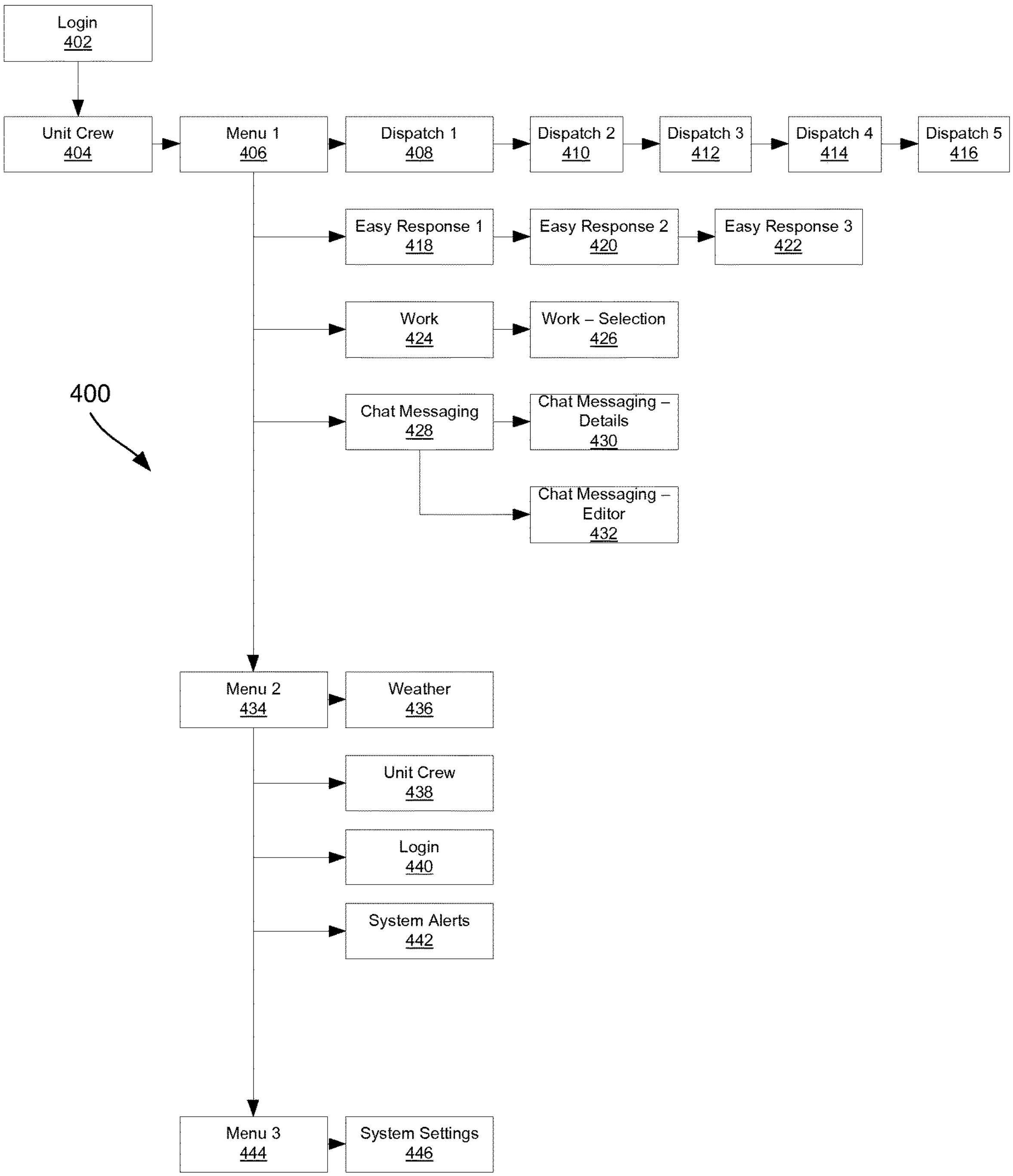
Login
402
Unit Crew
404
Menu 1
406
Dispatch 1
408
Dispatch 2
410
Dispatch 3
412
Dispatch 4
414
Dispatch 5
416
Easy Response 1
418
Easy Response 2
420
Easy Response 3
422
Work
424
Work – Selection
426
400
Chat Messaging
428
Chat Messaging – Details
430
Chat Messaging – Editor
432
Menu 2
434
Weather
436
Unit Crew
438
Login
440
System Alerts
442
Menu 3
444
System Settings
446

FIG. 4

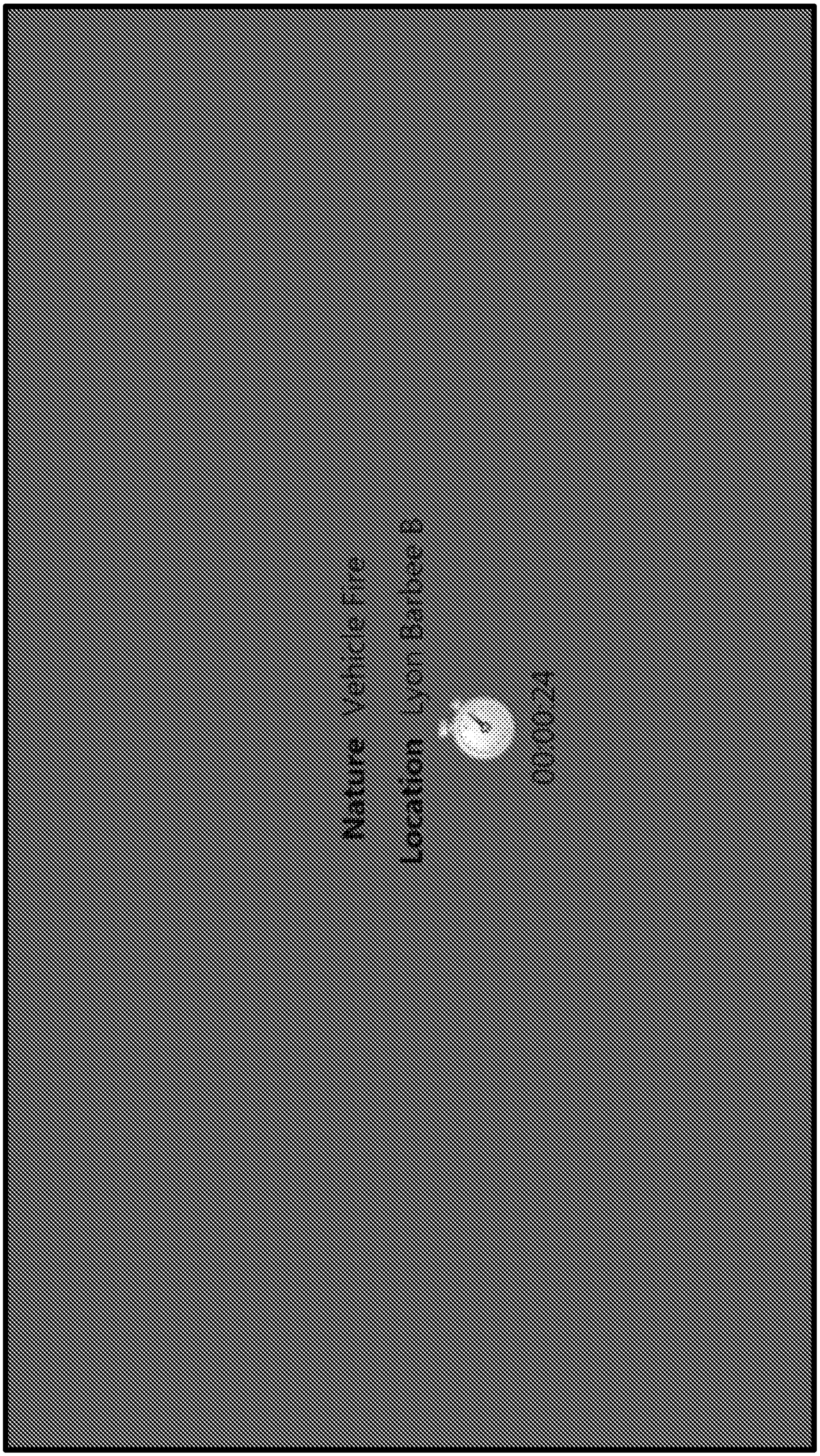

FIG. 5

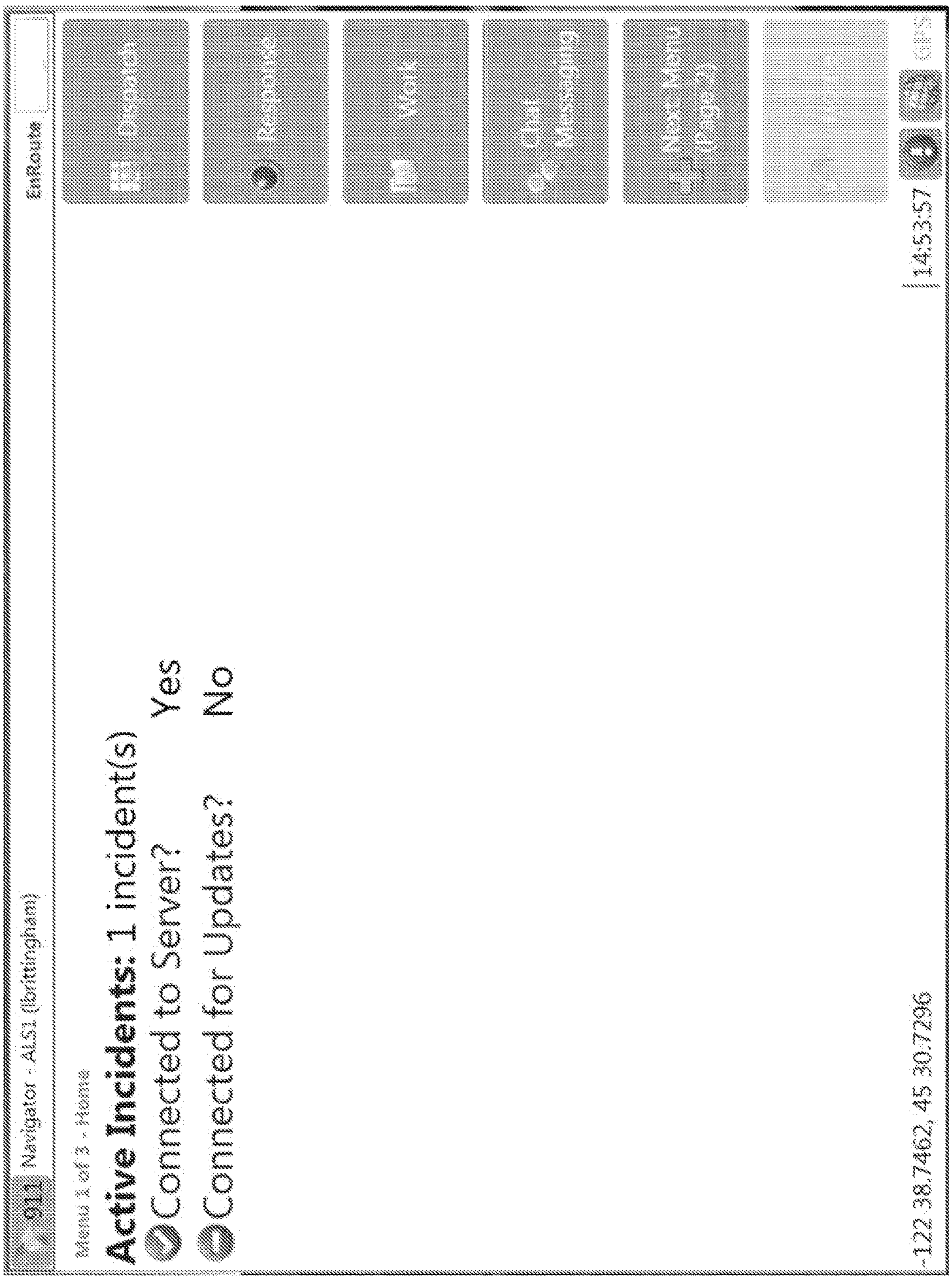
911
Navigator - ALS1 (lbrittingham)
EnRoute
Menu 1 of 3 - Home
Active Incidents: 1 incident(s)
Connected to Server? Yes
Connected for Updates? No
Dispatch
Response
Work
Chat Messaging
Next Menu (Page 2)
-122 38.7462, 45 30.7296
14:53:57
GPS

FIG. 6

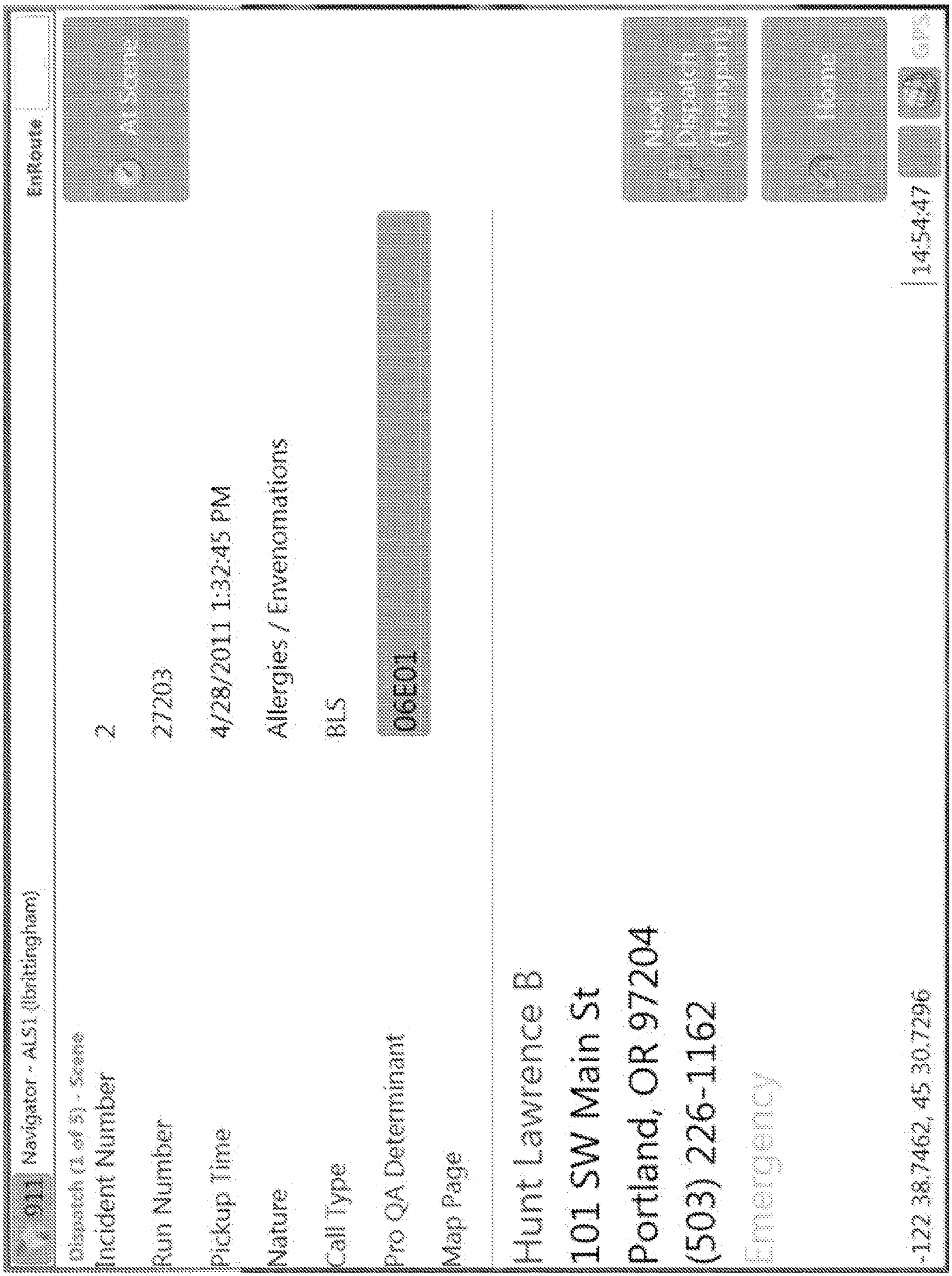
911
Navigator - ALS1 (brittingham)
EnRoute
Dispatch (1 of 5) - Scene
Incident Number
2
Run Number
27203
Pickup Time
4/28/2011 1:32:45 PM
Nature
Allergies / Envenomations
Call Type
BLS
Pro QA Determinant
06E01
Map Page
At Scene
Hunt Lawrence B
101 SW Main St
Portland, OR 97204
(503) 226-1162
Emergency
Next Dispatch (Transport)
Home
-122 38.7462, 45 30.7296
14:54:47
GPS

FIG. 7

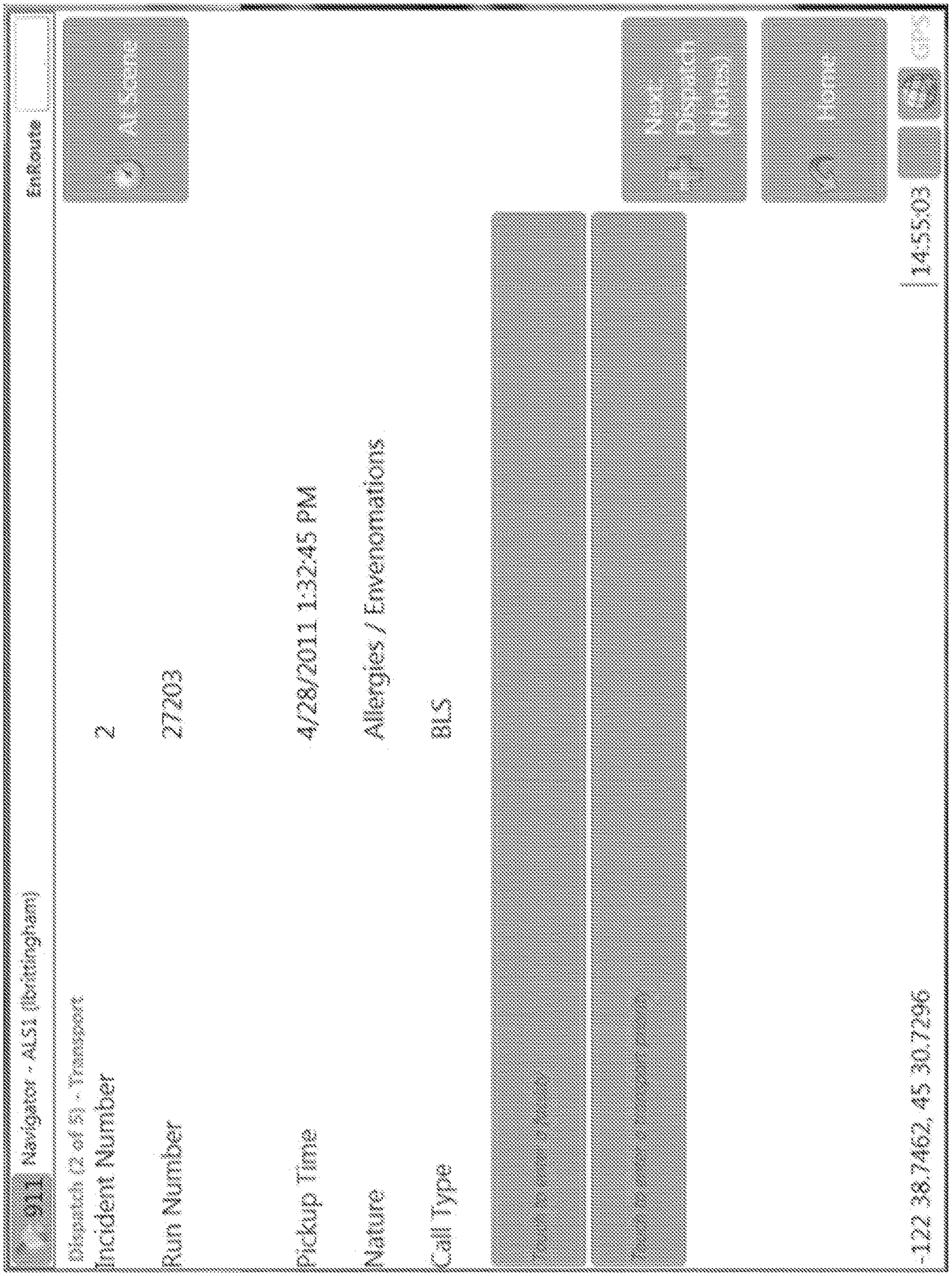
911
Navigator - ALS1 (lbrittingham)
EnRoute
At Scene
Dispatch (2 of 5) - Transport
Incident Number
2
Run Number
27203
Pickup Time
4/28/2011 1:32:45 PM
Nature
Allergies / Envenomations
Call Type
BLS
Next Dispatch (Notes)
Home
-122 38.7462, 45 30.7296
14:55:03
GPS

FIG. 8

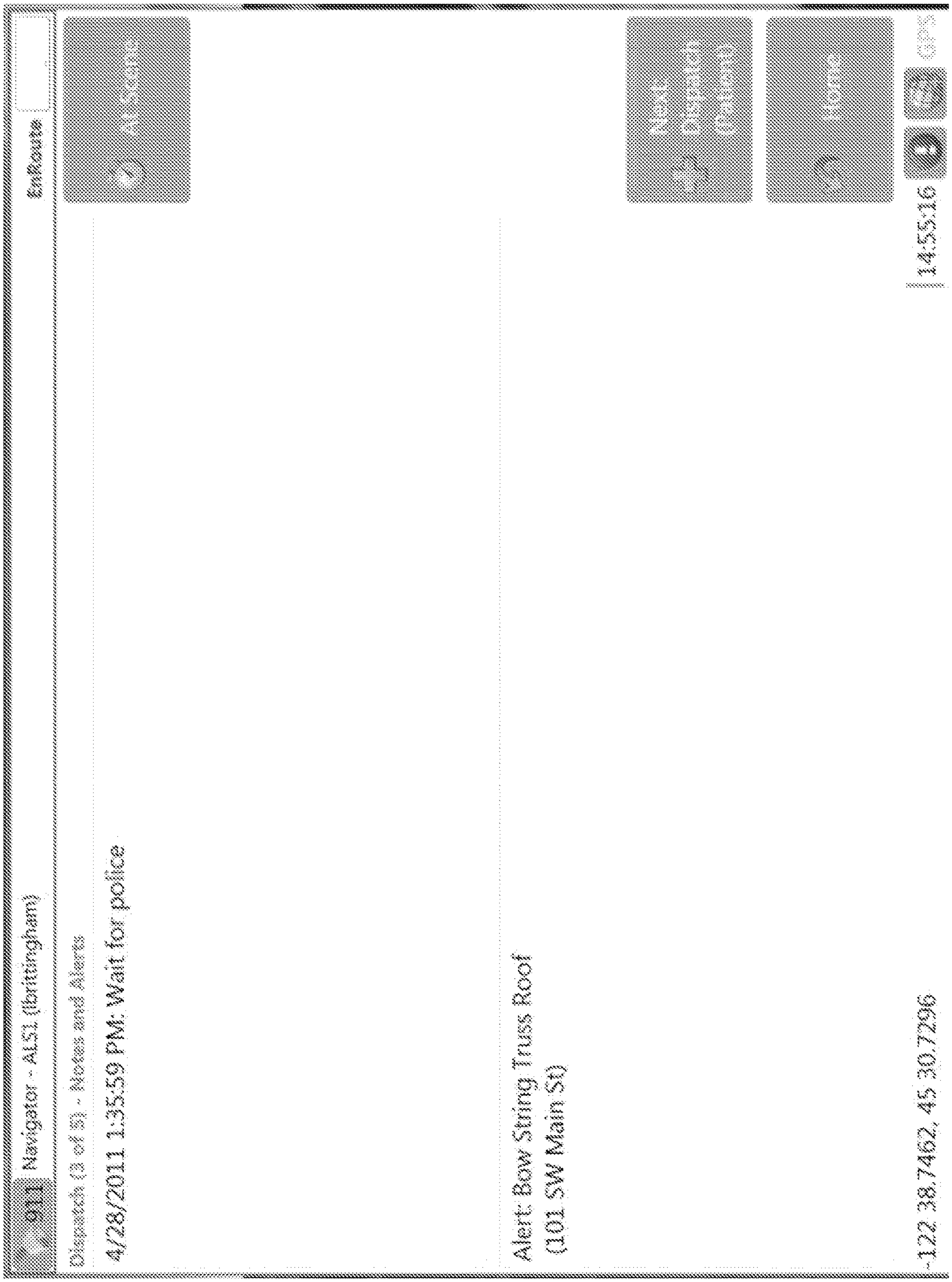
911
Navigator - ALS1 (lbrittingham)
EnRoute
Dispatch (3 of 5) - Notes and Alerts
4/28/2011 1:35:59 PM: Wait for police
At Scene
Alert: Bow String Truss Roof
(101 SW Main St)
Next Dispatch (Patient)
Home
-122 38.7462, 45 30.7296
14:55:16
GPS

FIG. 9

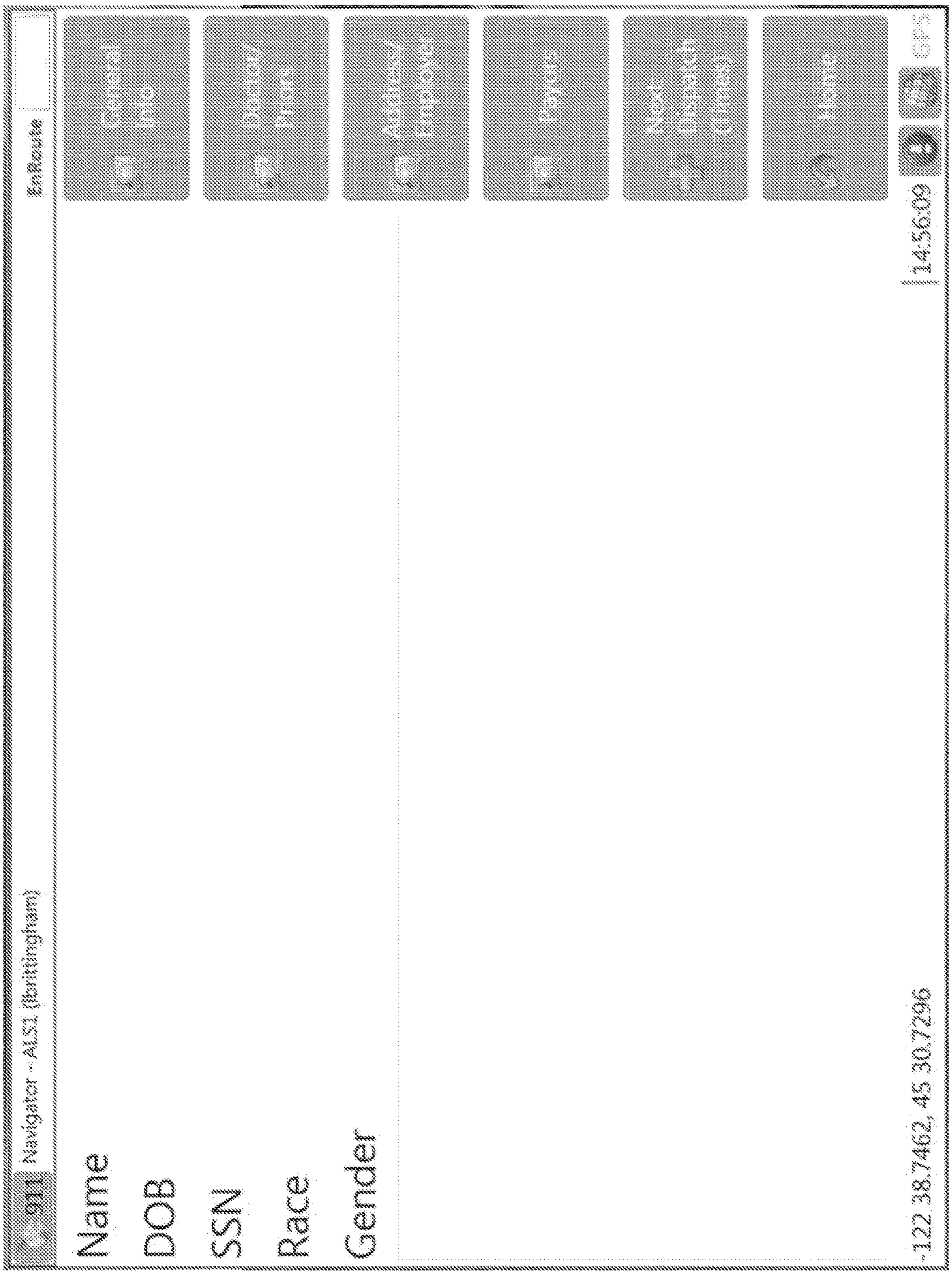
911
Navigator - ALS1 (Brittingham)
EnRoute
Name
DOB
SSN
Race
Gender
General Info
Address/ Employer
Next Dispatch (Times)
Home
14:56:09
GPS
-122 38.7462, 45 30.7296

FIG. 10

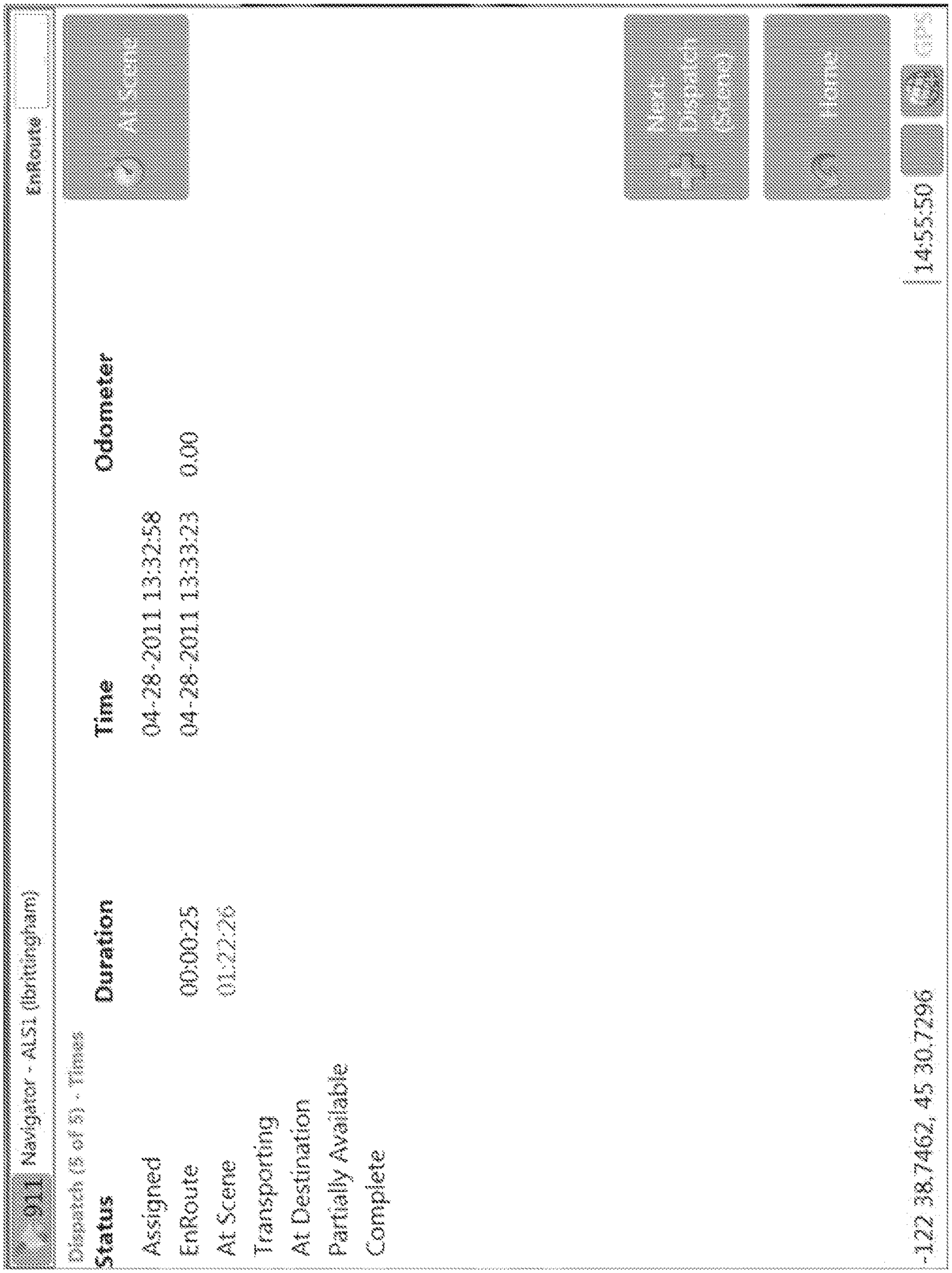
911
Navigator - ALS1 (lbrittingham)
EnRoute
Dispatch (5 of 5) - Times
Status
Duration
Time
Odometer
Assigned
04-28-2011 13:32:58
EnRoute
00:00:25
04-28-2011 13:33:23
0.00
At Scene
01:22:26
Transporting
At Destination
Partially Available
Complete
At Scene
Next Dispatch (Scene)
Home
-122 38.7462, 45 30.7296
14:55:50
GPS

FIG. 11

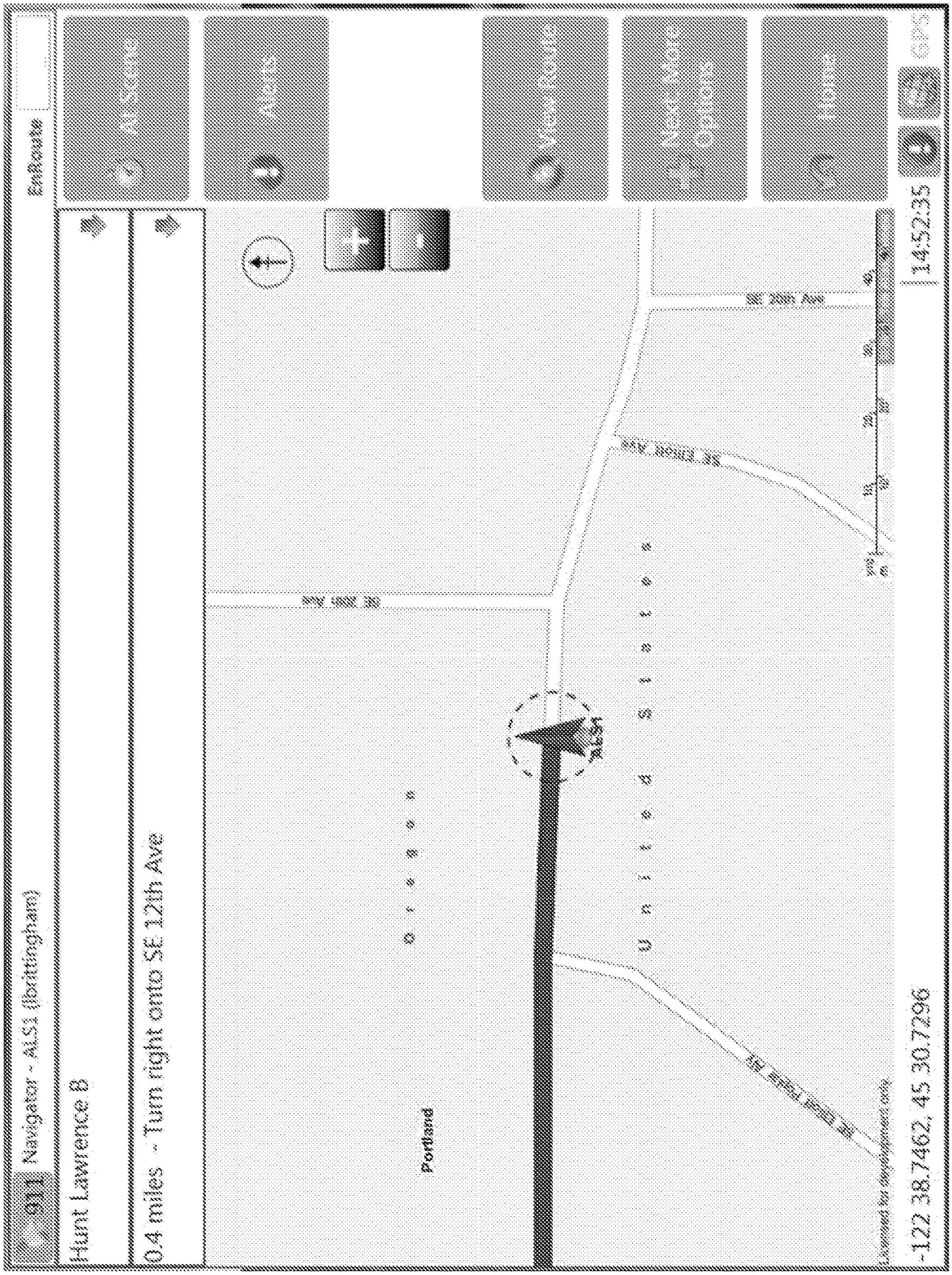
911
Navigator - ALS1 (jbrittingham)
EnRoute
Hunt Lawrence B
0.4 miles - Turn right onto SE 12th Ave
At Scene
Alerts
View Route
Next More Options
Home
Portland
Oregon
United States
ALS1
SE 20th Ave
14:52:35
GPS
-122 38.7462, 45 30.7296

FIG. 12

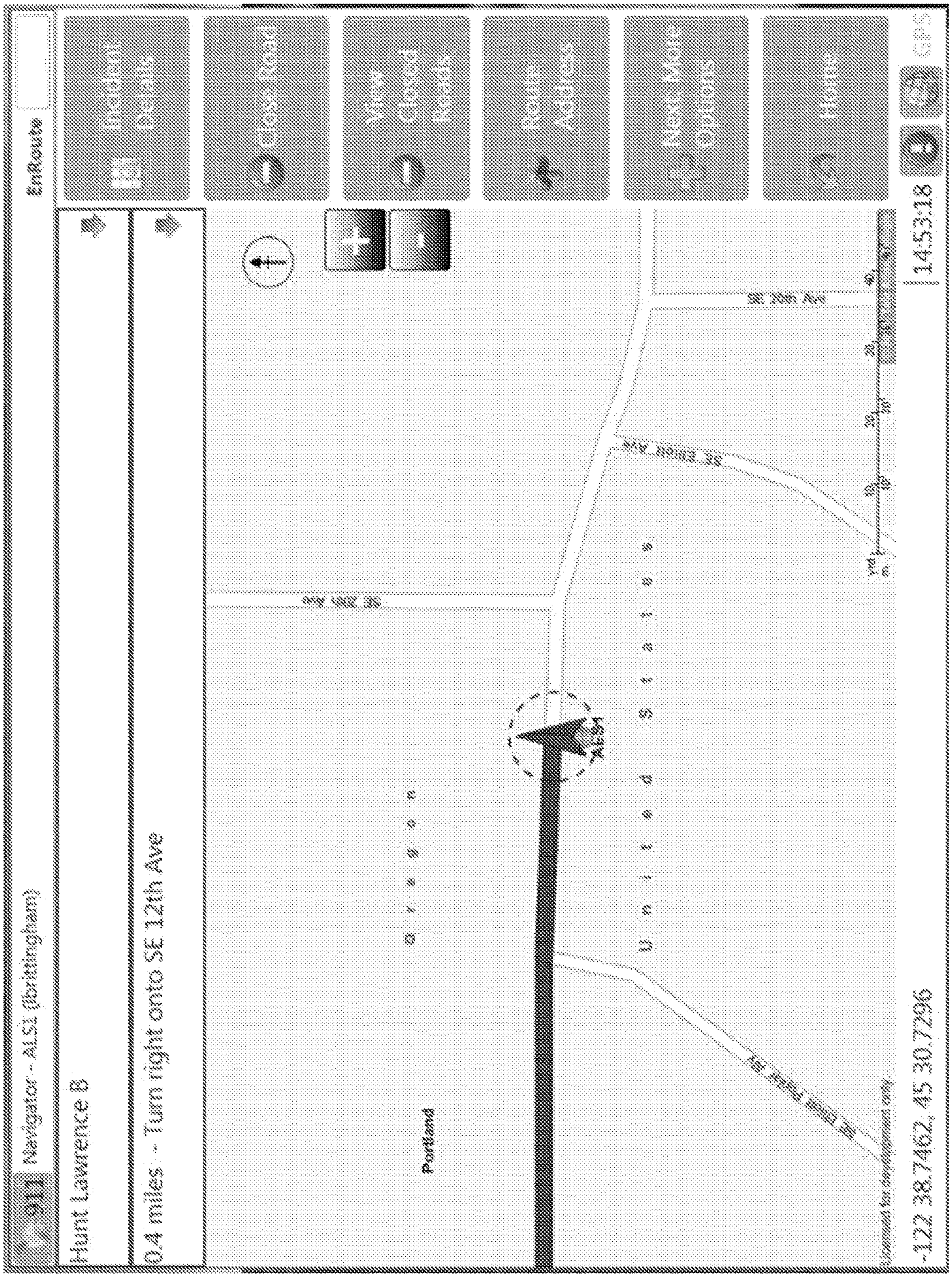
911
Navigator - ALS1 (jbrittingham)
EnRoute
Hunt Lawrence B
0.4 miles - Turn right onto SE 12th Ave
Incident Details
Close Road
View Closed Roads
Route Address
Next More Options
Home
GPS
14:53:18
Portland
Oregon
United States
SE 20th Ave
ALS1
-122 38.7462, 45 30.7296

FIG. 13

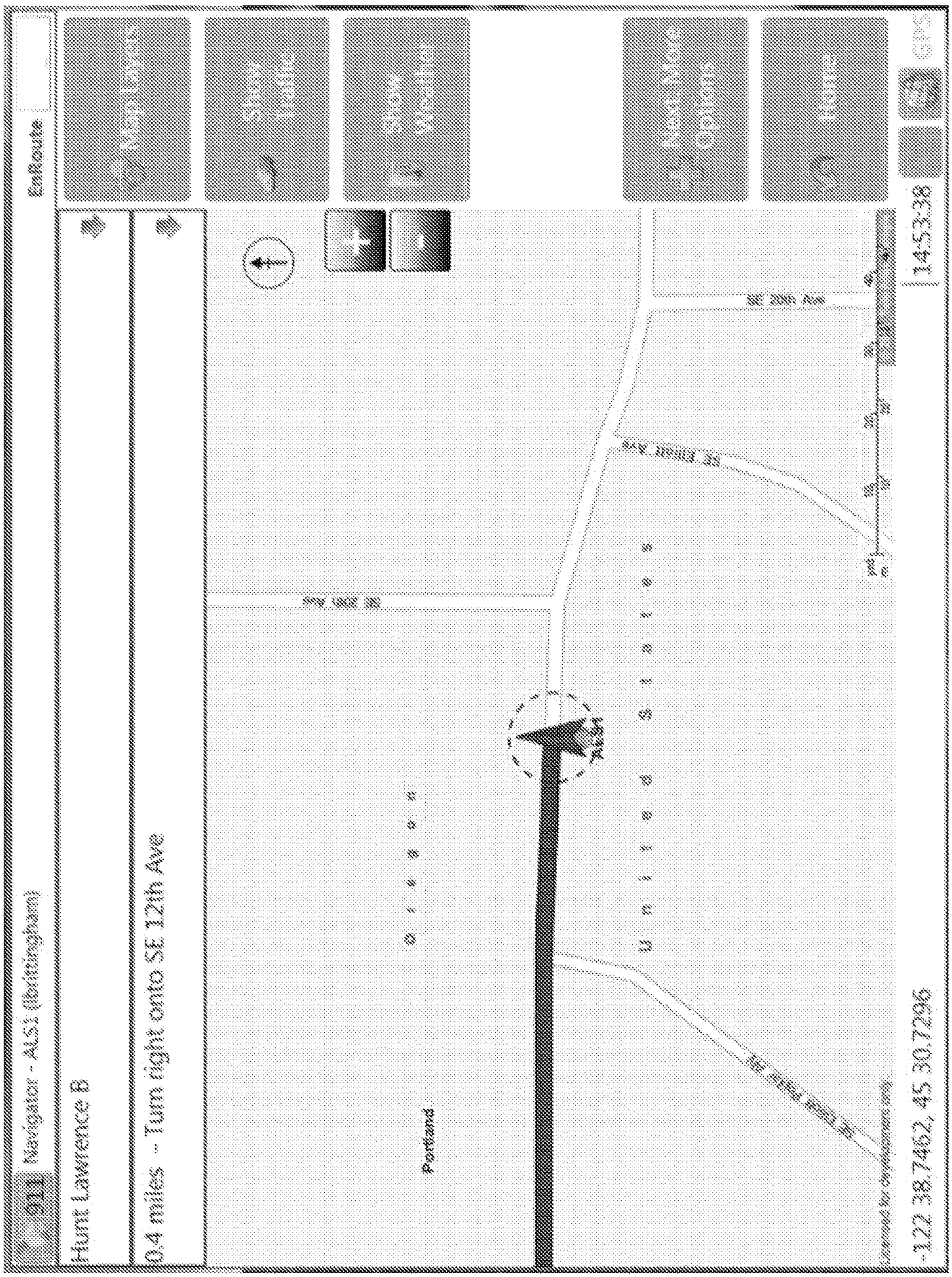
911
Navigator - ALS1 (lbrittingham)
EnRoute
Hunt Lawrence B
0.4 miles - Turn right onto SE 12th Ave
Map layers
Show Traffic
Show Weather
Next More Options
Home
GPS
SE 20th Ave
Portland
ALS1
14:53:38
-122 38.7462, 45 30.7296

FIG. 14

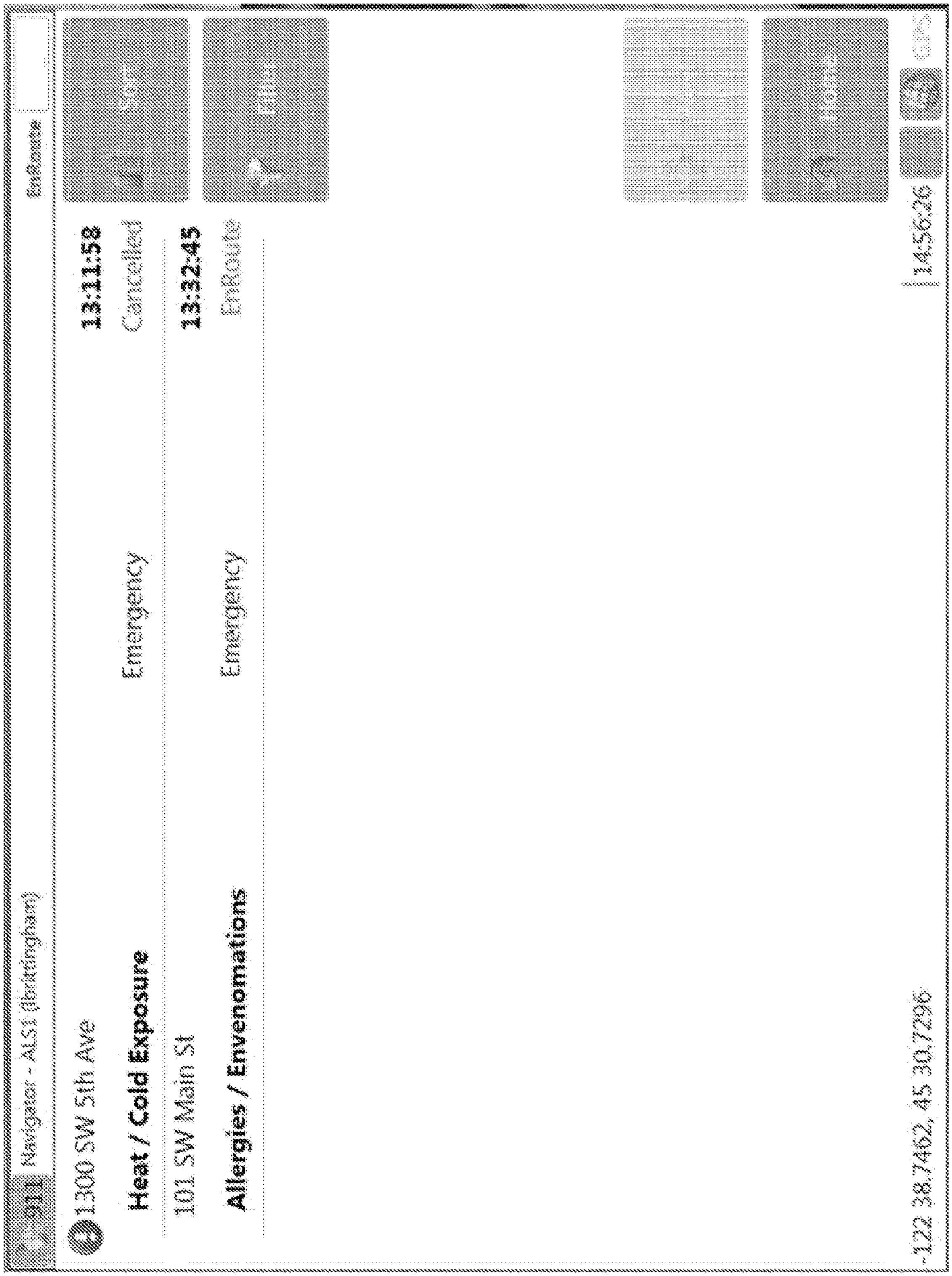
911
Navigator - ALS1 (lbrittingham)
EnRoute
1300 SW 5th Ave
Heat / Cold Exposure
Emergency
13:11:58
Cancelled
101 SW Main St
Allergies / Envenomations
Emergency
13:32:45
EnRoute
Sort
Filter
Home
-122 38.7462, 45 30.7296
14:56:26
GPS

FIG. 15

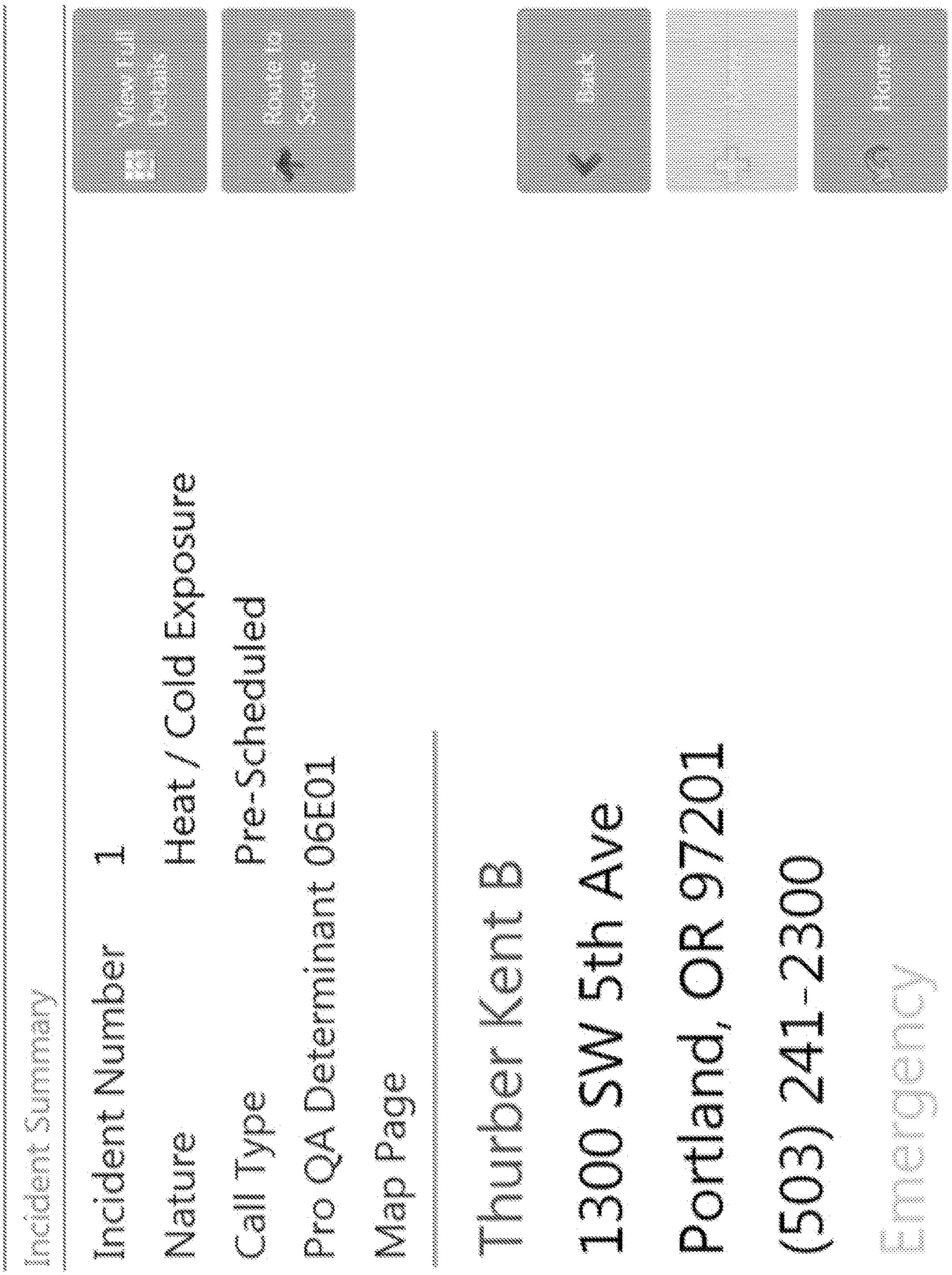
Incident Summary
Incident Number 1
Nature Heat / Cold Exposure
Call Type Pre-Scheduled
Pro QA Determinant 06E01
Map Page
Thurber Kent B
1300 SW 5th Ave
Portland, OR 97201
(503) 241-2300
Emergency
View Full Details
Route to Scene
Back
Home

FIG. 16

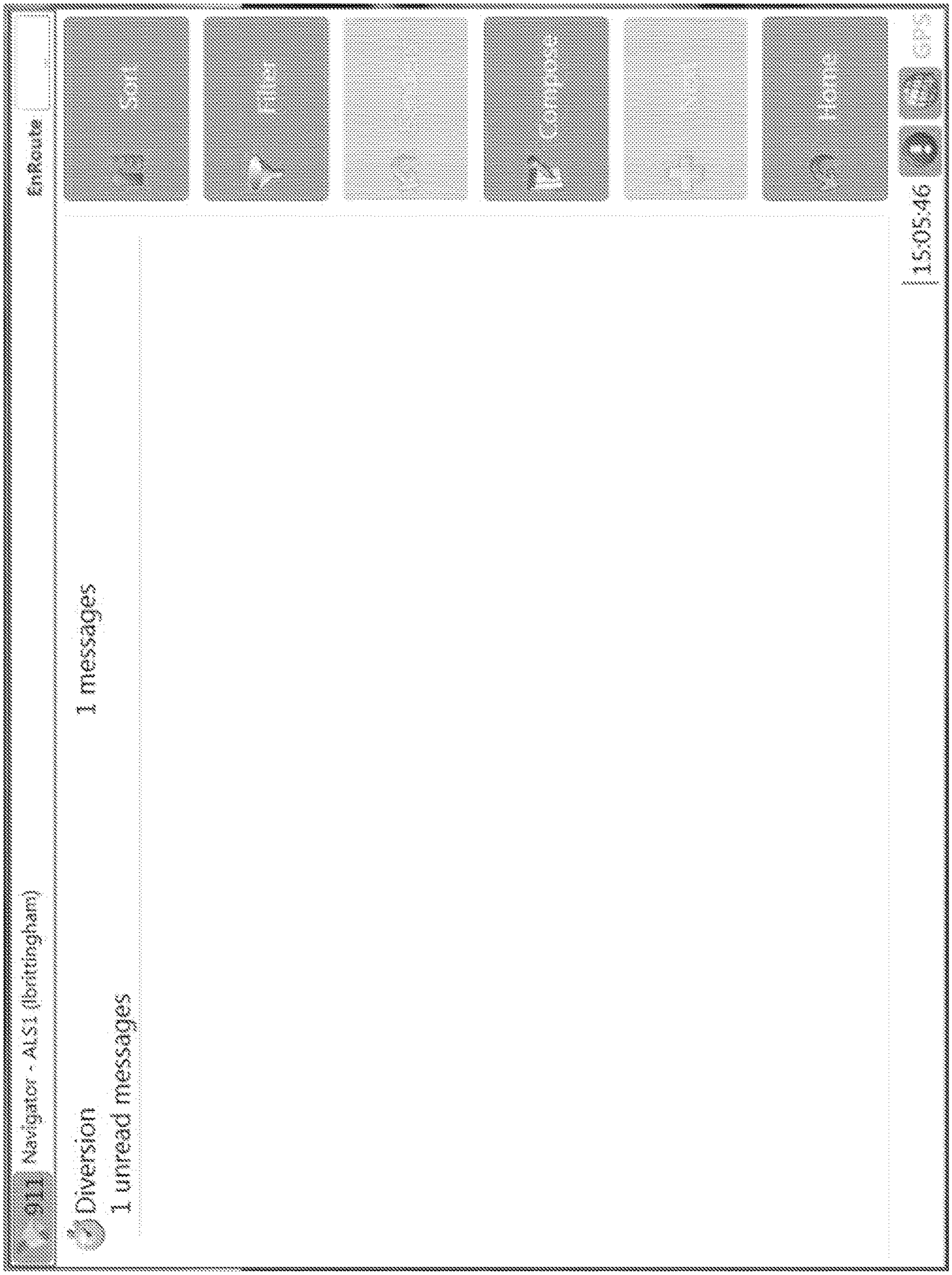
911
Navigator - ALS1 (jbrittingham)
EnRoute
Diversion
1 unread messages
1 messages
Sort
Filter
Compose
Home
15:05:46
GPS

FIG. 17

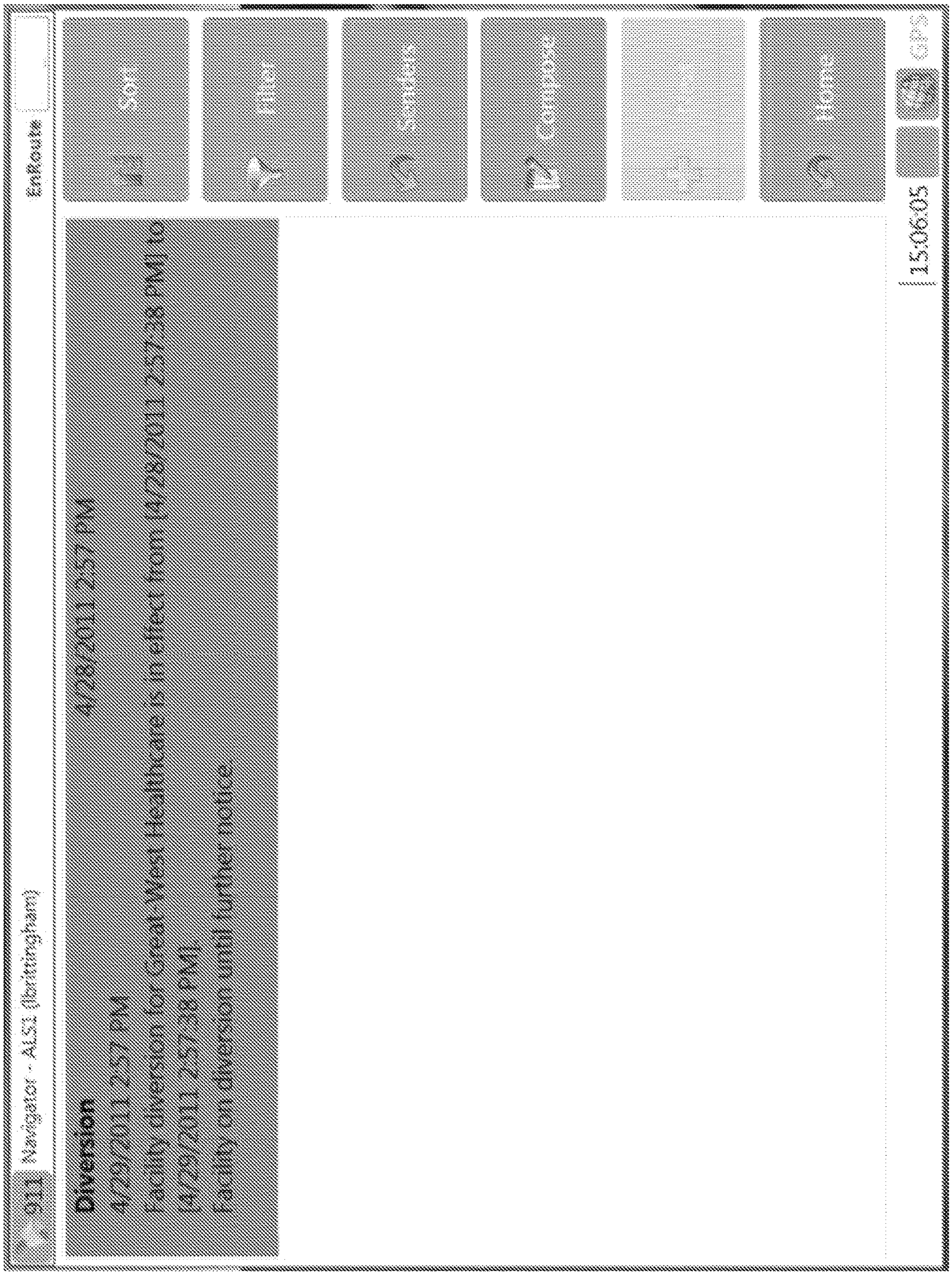
911
Navigator - ALS1 (lbrittingham)
EnRoute
Diversion
4/29/2011 2:57 PM
Facility diversion for Great-West Healthcare is in effect from [4/28/2011 2:57:38 PM] to [4/29/2011 2:57:38 PM].
Facility on diversion until further notice.
4/28/2011 2:57 PM
Sort
Filter
Senders
Compose
Home
15:06:05
GPS

FIG. 18

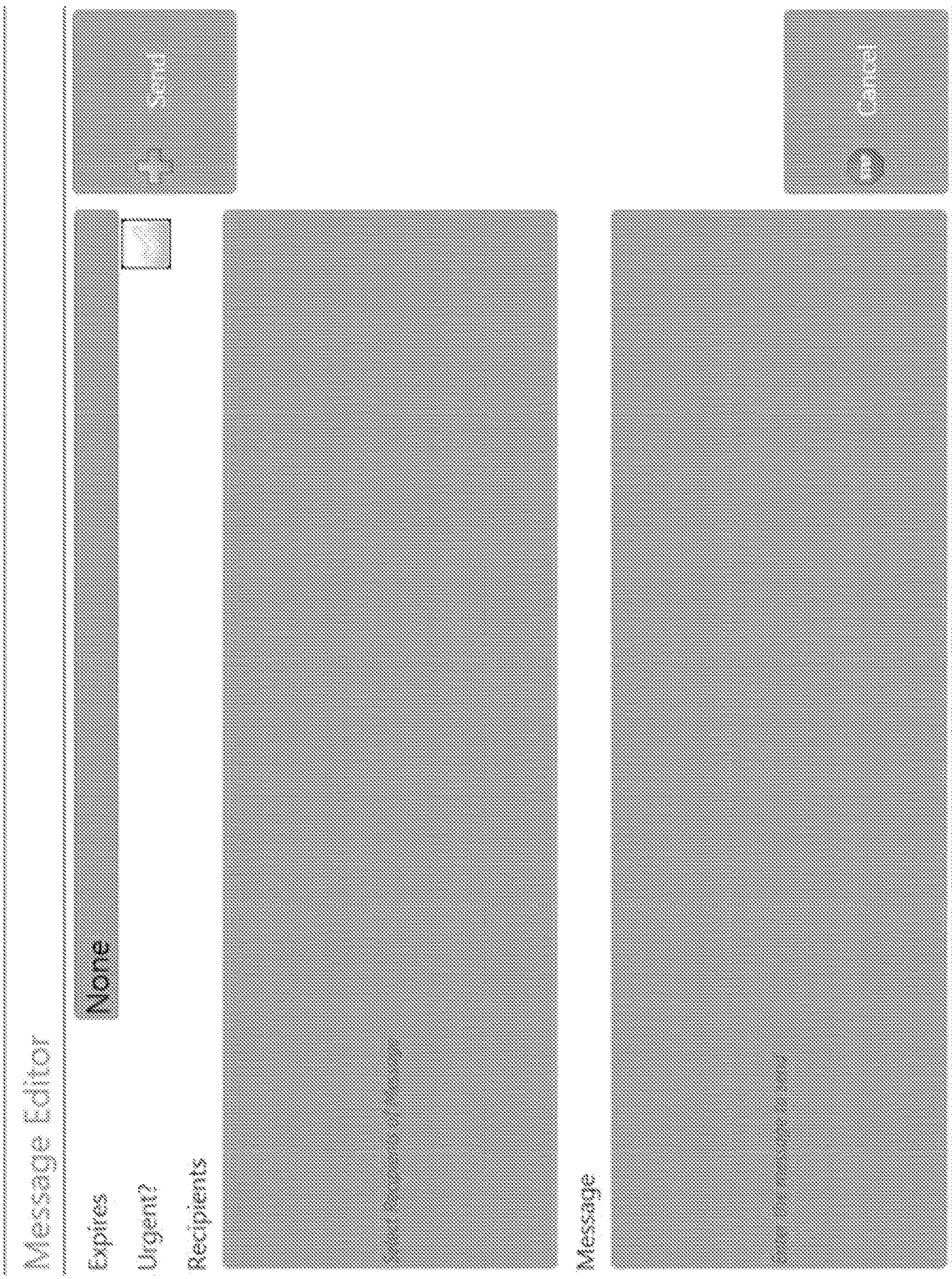
Message Editor
Expires
None
Urgent?
Recipients
Message
Send
Cancel

FIG. 19

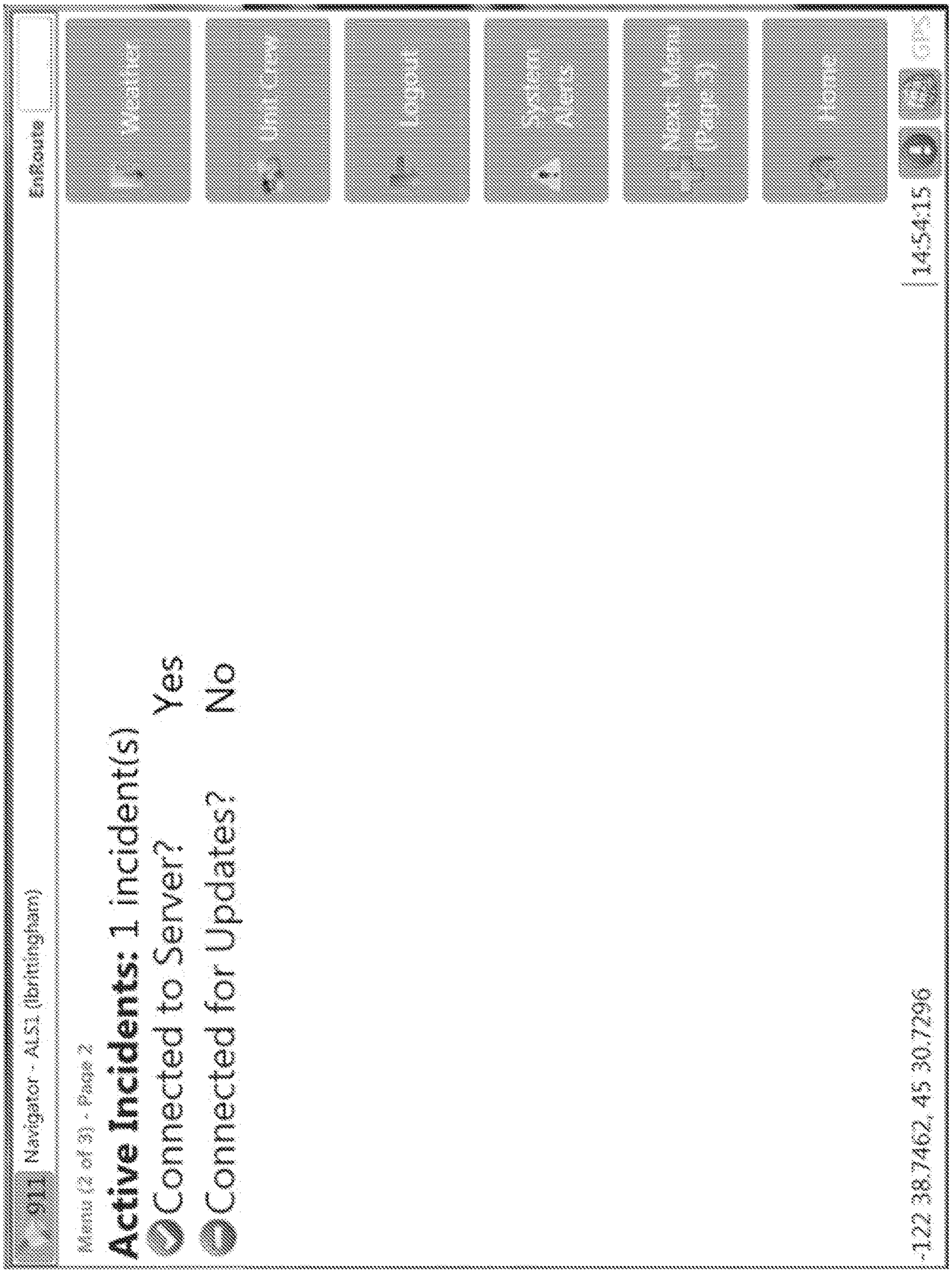
911
Navigator - ALS1 (lbrittingham)
EnRoute
Menu (2 of 3) - Page 2
Active Incidents: 1 incident(s)
Connected to Server? Yes
Connected for Updates? No
Weather
Unit Crew
Logout
System Alerts
Next Menu (Page 3)
Home
-122 38.7462, 45 30.7296
14:54:15
GPS

FIG. 20


911
Navigator - ALS1 (lbrittingham)
EnRoute
64 °F
Feels Like 64 °F
Updated 4/28/11 12:47 PM MDT
UV Index
9 Very High
Wind
From ESE at 6 mph
Humidity
6%
Pressure
30.04 in Falling
Dew Point
N/A
Visibility
N/A
Today
67 °F
Precipitation 0%
Wind WSW at 12 mph
Partly Cloudy
Tonight
44 °F
Precipitation 10%
Wind WSW at 10 mph
Sunset: 7:52 PM
Partly Cloudy
-122 38.7462, 45 30.7296
14:59:36
Home
GPS

FIG. 21

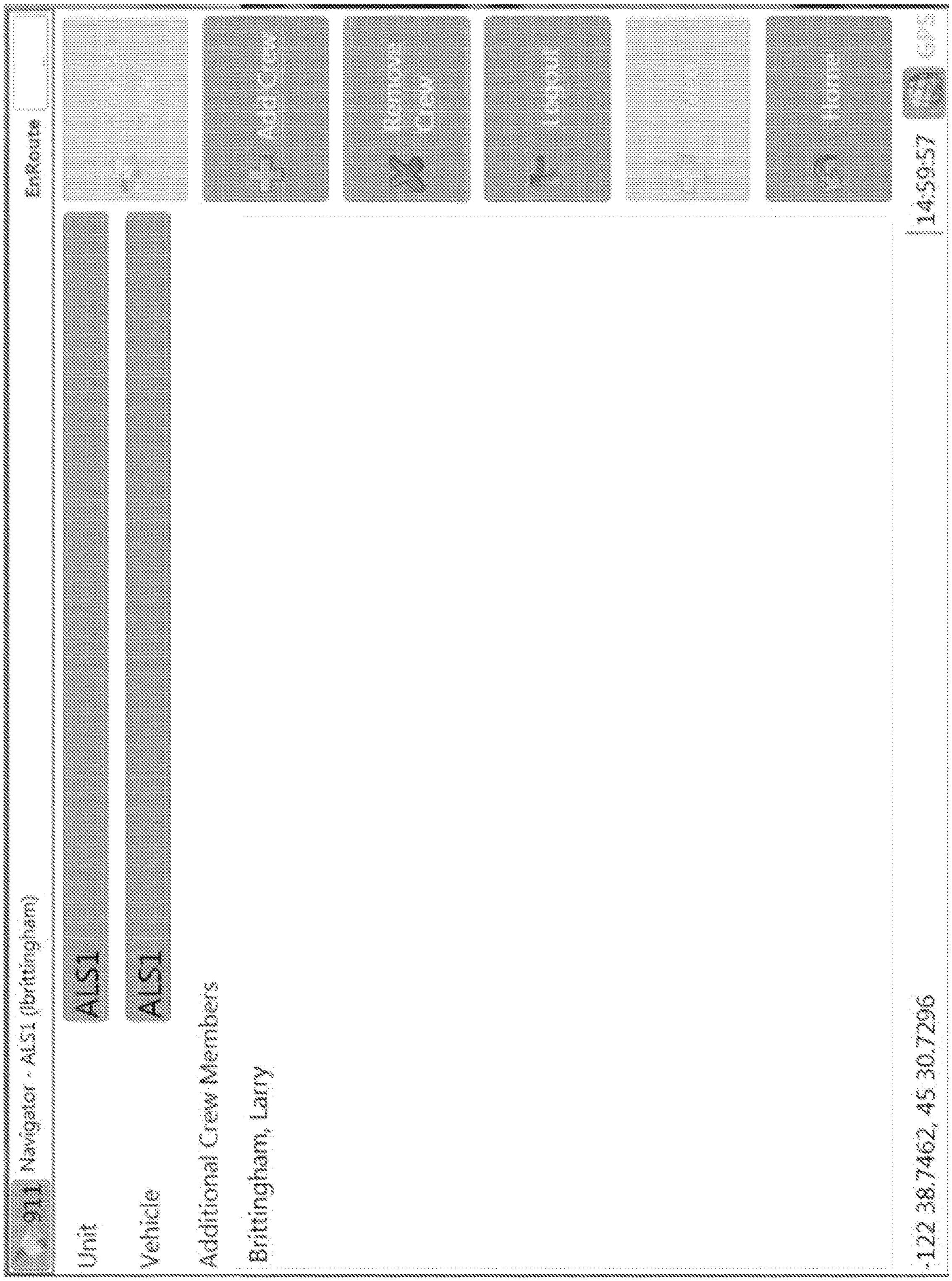
911
Navigator - ALS1 (lbrittingham)
EnRoute
Unit
ALS1
Vehicle
ALS1
Additional Crew Members
Brittingham, Larry
Add Crew
Remove Crew
Logout
Home
-122 38.7462, 45 30.7296
14:59:57
GPS

FIG. 22

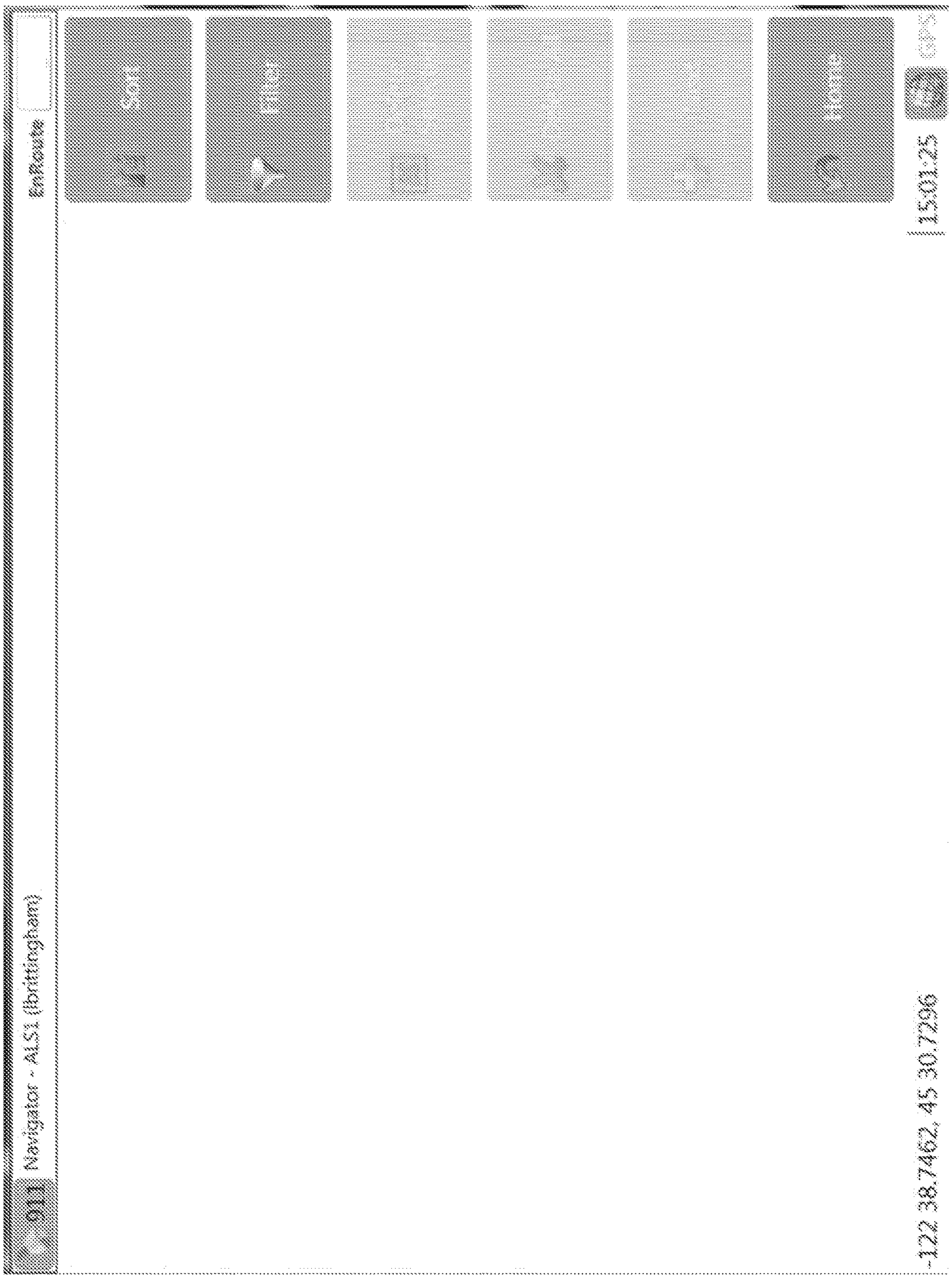
911
Navigator - ALS1 (lbrittingham)
EnRoute
Sort
Filter
Home
15:01:25
GPS
-122 38.7462, 45 30.7296

FIG. 23

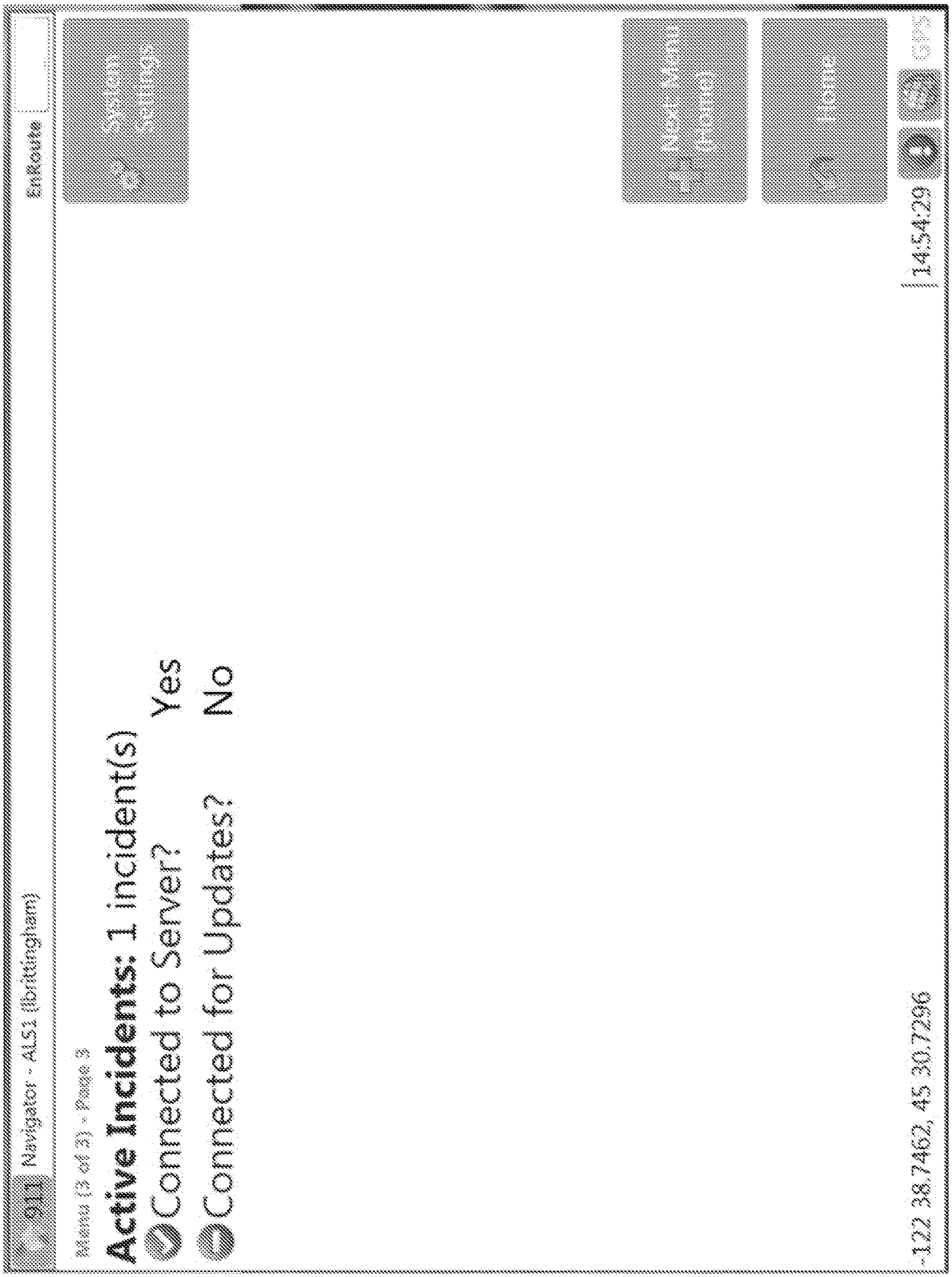
911
Navigator - ALS1 (lbrittingham)
EnRoute
Menu (3 of 3) - Page 3
Active Incidents: 1 incident(s)
Connected to Server? Yes
Connected for Updates? No
System Settings
Next Menu (Home)
Home
-122 38.7462, 45 30.7296
14:54:29
GPS

FIG. 24

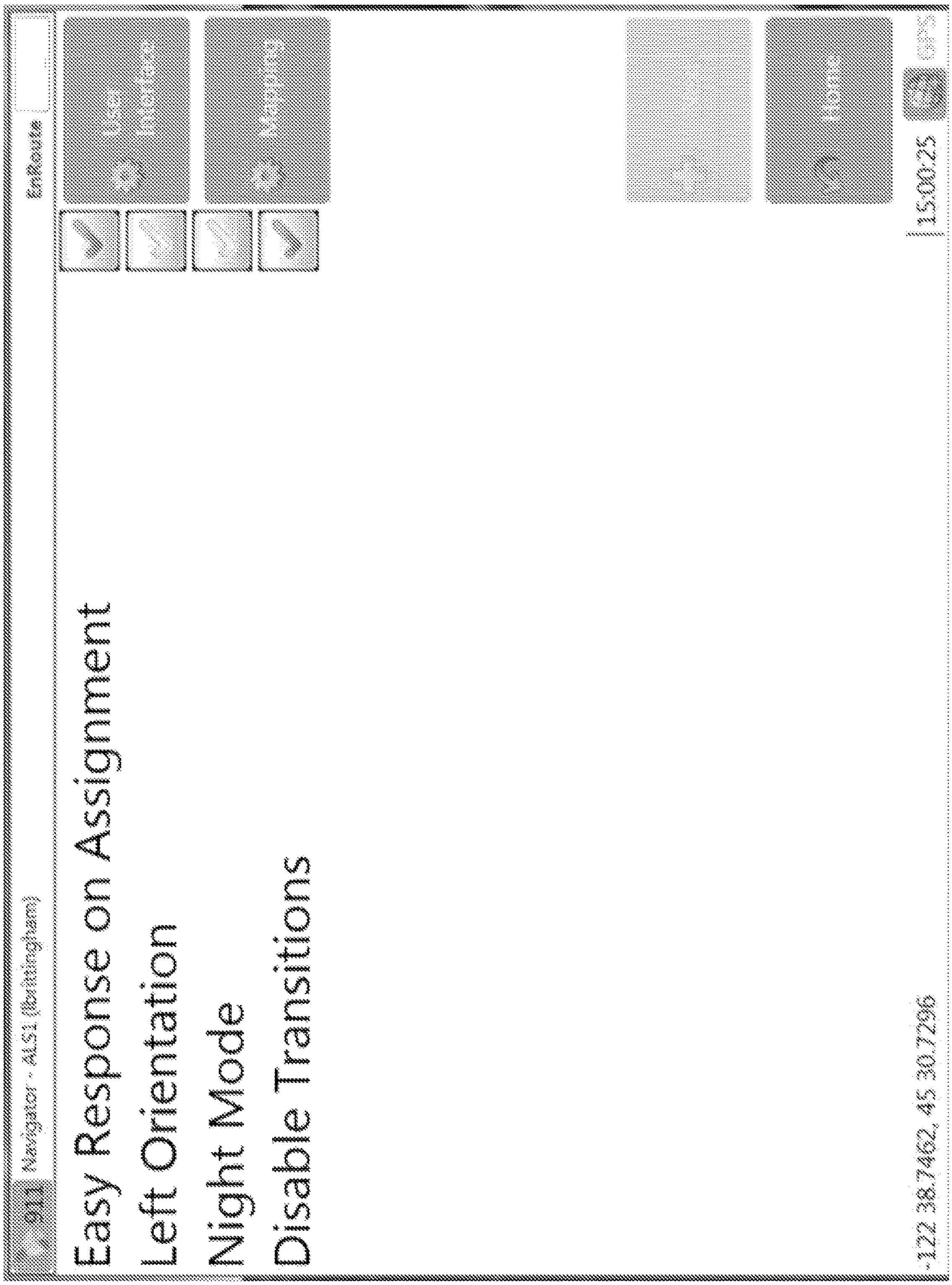
911
Navigator - ALS1 (fbrittingham)
EnRoute
Easy Response on Assignment
Left Orientation
Night Mode
Disable Transitions
User Interface
Mapping
Home
-122 38.7462, 45 30.7296
15:00:25
GPS

FIG. 25

SYSTEM FOR TRACKING AND MANAGEMENT OF EMS RESPONSES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims benefit under 35 U.S.C. § 120 as a continuation of U.S. Non-Provisional patent application Ser. No. 17/152,133, filed Jan. 19, 2021, which claims benefit under 35 U.S.C. § 120 as a continuation of U.S. Non-Provisional patent application Ser. No. 16/786,203, filed Feb. 10, 2020 (now U.S. Pat. No. 10,942,040), which claims benefit under 35 U.S.C. § 120 as a continuation of U.S. Non-Provisional patent application Ser. No. 13/467,859, filed May 9, 2012 (now U.S. Pat. No. 10,598,508), which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 61/484,121, filed on May 9, 2011, each of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

Embodiments of the present invention relate generally to user interface systems, and more particularly to user interface systems for emergency medical services (EMS) navigation systems.

BACKGROUND

Current user interface systems for EMS devices typically include rather small navigation buttons spread out in various locations on the user's screen. For workers in the EMS field, particularly workers who interact with patients in a mobile environment, such complicated user interface setups with small buttons often require the workers to devote larger amounts of time to finding the correct button, and/or often divert the workers' attention or vision from the EMS situation (e.g. patient care or driving conditions) to focus on navigating the user interface. For example, an ambulance driver who is using an EMS navigation system may need to stop the vehicle in order to divert the necessary attention to activating navigation commands in an EMS navigation device. EMS navigation interfaces which include small buttons spread out in various locations on the user's screen detract the EMS technician's attention from other more valuable targets during an EMS response event.

SUMMARY

A method for interfacing with a user of an EMS navigation system according to embodiments of the present invention includes displaying on a device screen a series of display screens that are navigationally linked by a series of navigation buttons, wherein the series of display screens comprises a home screen, wherein each display screen of the series of display screens comprises a home button linking to the home screen, wherein each navigation button of the series of navigation buttons appearing in a display screen of the series of display screens is substantially the same size and larger than other buttons in the display screen, and wherein each navigation button of the series of navigation buttons appearing in the display screen of the series of display screens is adjacent to a left edge or a right edge of the device screen to facilitate use by EMS users.

A method for interfacing with a user of an EMS navigation system according to embodiments of the present invention includes displaying on a device screen a series of

display screens that are navigationally linked by a series of navigation buttons, wherein the series of display screens comprises a home screen, wherein each display screen of the series of display screens comprises a home button linking to the home screen, wherein each navigation button of the series of navigation buttons appearing in a display screen of the series of display screens is larger than any other buttons in the display screen, and wherein each navigation button of the series of navigation buttons appearing on the display screen of the series of display screens is adjacent to a left edge or a right edge of the device screen. A user may toggle or cycle through two or more of the series of display screens by selecting a toggle button which has a same position on the device screen in each of the two or more of the series of display screens.

A method for interfacing with a user of an EMS navigation system according to embodiments of the present invention includes receiving an incident assignment from a dispatcher corresponding to an EMS incident, displaying an easy response button with a device touchscreen, wherein the easy response button has an area substantially as large as the device touchscreen, detecting a user's touch on the easy response button on the device touchscreen, and based on the detection of the user's touch, updating a status of the EMS navigation system to indicate that response is being made.

The method as described above, further including, based on the detection of the user's touch, automatically displaying a map with a route to a target destination corresponding to the EMS incident.

The methods as described above, wherein updating a status of the EMS navigation system to indicate that response is being made includes starting a response timer.

The methods as described above, wherein displaying the easy response button includes displaying a timer indicating a time since the incident assignment was received from the dispatcher.

The methods as described above, wherein displaying the easy response button includes displaying a nature of the EMS incident.

The methods as described above, wherein displaying the easy response button includes displaying a location of the EMS incident.

A system for EMS navigation according to embodiments of the present invention includes a touch screen display device, the touch screen display device configured to display information related to EMS navigation and to accept tactile input from a user, the touch screen configured to display a series of display screens that are navigationally linked by a series of navigation buttons, wherein the series of display screens comprises a home screen, wherein each display screen of the series of display screens comprises a home button linking to the home screen, wherein each navigation button of the series of navigation buttons appearing in a display screen of the series of display screens is larger than any other buttons on the display screen, and wherein each navigation button of the series of navigation buttons appearing in the display screen of the series of display screens is adjacent to a left edge or a right edge of the device screen.

While multiple embodiments are disclosed, still other embodiments of the present invention will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates a system for mobile and enterprise user real-time display of medical information collected from multiple different EMS devices, according to embodiments of the present invention.

FIG. **2** illustrates an exemplary computer system, according to embodiments of the present invention.

FIG. **3** depicts a flow diagram illustrating an activation of response status through use of an easy response button, according to embodiments of the present invention.

FIG. **4** illustrates a navigational linking relationship between display screens of an EMS navigation device, according to embodiments of the present invention.

FIG. **5** illustrates an easy response button display screen, according to embodiments of the present invention.

FIG. **6** illustrates a home screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **7** illustrates a first dispatch screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **8** illustrates a second dispatch screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **9** illustrates a third dispatch screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **10** illustrates a fourth dispatch screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **11** illustrates a fifth dispatch screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **12** illustrates a first easy response screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **13** illustrates a second easy response screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **14** illustrates a third easy response screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **15** illustrates a work screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **16** illustrates a work selection screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **17** illustrates a chat messaging screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **18** illustrates a chat messaging details screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **19** illustrates a chat messaging editor screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **20** illustrates a second menu screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **21** illustrates a weather screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **22** illustrates a unit crew screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **23** illustrates a system alerts screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **24** illustrates a third menu screen of an EMS navigation system, according to embodiments of the present invention.

FIG. **25** illustrates a system settings screen of an EMS navigation system, according to embodiments of the present invention.

While the invention is amenable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the invention to the particular embodiments described. On the contrary, the invention is intended to cover all modifications, equivalents, and alternatives falling within the scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

As illustrated in FIG. **1**, a system **100** according to embodiments of the present invention performs advanced data management, integration and presentation of EMS data from multiple different devices. System **100** includes a mobile environment **101**, an enterprise environment **102**, and an administration environment **103**. Devices within the various environments **101**, **102**, **103** may be communicably coupled via a network **120**, such as, for example, the Internet. System **100** is further described in Patent Cooperation Treaty Application Publication No. WO 2011/011454, published on Jan. 27, 2011, which is incorporated herein by reference in its entirety for all purposes.

As used herein, the phrase "communicably coupled" is used in its broadest sense to refer to any coupling whereby information may be passed. Thus, for example, communicably coupled includes electrically coupled by, for example, a wire; optically coupled by, for example, an optical cable; and/or wirelessly coupled by, for example, a radio frequency or other transmission media. "Communicably coupled" also includes, for example, indirect coupling, such as through a network, or direct coupling.

According to embodiments of the present invention, the mobile environment **101** is an ambulance or other EMS vehicle—for example a vehicular mobile environment (VME). The mobile environment may also be the local network of data entry devices as well as diagnostic and therapeutic devices established at time of treatment of a patient or patients in the field environment—the "At Scene Patient Mobile Environment" (ASPME). The mobile environment may also be a combination of one or more of VMEs and/or ASPMEs. The mobile environment may include a navigation device **110** used by the driver **112** to track the mobile environment's position **101**, locate the mobile environment **101** and/or the emergency location, and locate the transport destination, according to embodiments of the present invention. The navigation device **110** may include a Global Positioning System ("GPS"), for example. The navigation device **110** may also be configured to perform calculations about vehicle speed, the travel time between locations, and estimated times of arrival. According to embodiments of the present invention, the navigation device **110** is located at the front of the ambulance to assist the driver **112** in navigating the vehicle. The navigation device **110** may be, for example, a RescueNet® Navigator onboard electronic data communication system available from Zoll Data Systems of Broomfield, Colorado.

As illustrated in FIG. **1**, a patient monitoring device **106** and a patient charting device **108** are also often used for patient care in the mobile environment **101**, according to embodiments of the present invention. The EMS technician

114 attaches the patient monitoring device **106** to the patient **116** to monitor the patient **116**. The patient monitoring device **106** may be, for example, a defibrillator device with electrodes and/or sensors configured for attachment to the patient **116** to monitor heart rate and/or to generate electrocardiographs ("ECG's"), according to embodiments of the present invention. The patient monitoring device **106** may also include sensors to detect or a processor to derive or calculate other patient conditions. For example, the patient monitoring device **106** may monitor, detect, treat and/or derive or calculate blood pressure, temperature, respiration rate, blood oxygen level, end-tidal carbon dioxide level, pulmonary function, blood glucose level, and/or weight, according to embodiments of the present invention. The patient monitoring device **106** may be a Zoll E-Series® defibrillator available from Zoll Medical Corporation of Chelmsford, Massachusetts, according to embodiments of the present invention. A patient monitoring device may also be a patient treatment device, or another kind of device that includes patient monitoring and/or patient treatment capabilities, according to embodiments of the present invention.

The patient charting device **108** is a device used by the EMS technician **114** to generate records and/or notes about the patient's **116** condition and/or treatments applied to the patient, according to embodiments of the present invention. For example, the patient charting device **108** may be used to note a dosage of medicine given to the patient **116** at a particular time. The patient charting device **108** and/or patient monitoring device **106** may have a clock, which may be synchronized with an external time source such as a network or a satellite to prevent the EMS technician from having to manually enter a time of treatment or observation (or having to attempt to estimate the time of treatment for charting purposes long after the treatment was administered), according to embodiments of the present invention. The patient charting device **108** may also be used to record biographic and/or demographic and/or historical information about a patient, for example the patient's name, identification number, height, weight, and/or medical history, according to embodiments of the present invention. According to embodiments of the present invention, the patient charting device **108** is a tablet PC, such as for example the TabletPCR component of the RescueNet® ePCR Suite available from Zoll Data Systems of Broomfield, Colorado. According to some embodiments of the present invention, the patient charting device **108** is a wristband or smartphone such as an Apple iPhone or iPad with interactive data entry interface such as a touch screen or voice recognition data entry that may be communicably connected to the BOA device **104** and tapped to indicate what was done with the patient **116** and when it was done.

The navigation device **110**, the charting device **108**, and the monitoring device **106** are each separately very useful to the EMS drivers **112** and technicians **114** before, during, and after the patient transport. A "back of ambulance" ("BOA") device **104** receives, organizes, stores, and displays data from each device **108**, **110**, **112** to further enhance the usefulness of each device **108**, **110**, **112** and to make it much easier for the EMS technician **114** to perform certain tasks that would normally require the EMS technician **114** to divert visual and manual attention to each device **108**, **110**, **112** separately, according to embodiments of the present invention. In other words, the BOA device centralizes and organizes information that would normally be de-centralized and disorganized, according to embodiments of the present invention.

The BOA device **104** is communicably coupled to the patient monitoring device **106**, the patient charting device **108**, and the navigation device **110**, according to embodiments of the present invention. The BOA device **104** is also communicably coupled to a storage medium **118**. The BOA device **104** may be a touch-screen, flat panel PC, and the storage medium **118** may be located within or external to the BOA device **104**, according to embodiments of the present invention. The BOA device **104** may include a display template serving as a graphical user interface, which permits the user (e.g. EMS tech **114**) to select different subsets and/or display modes of the information gathered from and/or sent to devices **106**, **108**, **110**, according to embodiments of the present invention.

Some embodiments of the present invention include various steps, some of which may be performed by hardware components or may be embodied in machine-executable instructions. These machine-executable instructions may be used to cause a general-purpose or a special-purpose processor programmed with the instructions to perform the steps. Alternatively, the steps may be performed by a combination of hardware, software, and/or firmware. In addition, some embodiments of the present invention may be performed or implemented, at least in part (e.g., one or more modules), on one or more computer systems, mainframes (e.g., IBM mainframes such as the IBM zSeries, Unisys ClearPath Mainframes, HP Integrity NonStop servers, NEC Express series, and others), or client-server type systems. In addition, specific hardware aspects of embodiments of the present invention may incorporate one or more of these systems, or portions thereof.

As such, FIG. **2** is an example of a computer system **200** with which embodiments of the present invention may be utilized. According to the present example, the computer system includes a bus **201**, at least one processor **202**, at least one communication port **203**, a main memory **204**, a removable storage media **205**, a read only memory **206**, and a mass storage **207**.

Processor(s) **202** can be any known processor, such as, but not limited to, an Intel® Itanium® or Itanium 2® processor(s), or AMD® Opteron® or Athlon MP® processor(s), or Motorola® lines of processors. Communication port(s) **203** can be any of an RS-232 port for use with a modem based dialup connection, a 10/100 Ethernet port, or a Gigabit port using copper or fiber, for example. Communication port(s) **203** may be chosen depending on a network such a Local Area Network (LAN), Wide Area Network (WAN), or any network to which the computer system **200** connects. Main memory **204** can be Random Access Memory (RAM), or any other dynamic storage device(s) commonly known to one of ordinary skill in the art. Read only memory **206** can be any static storage device(s) such as Programmable Read Only Memory (PROM) chips for storing static information such as instructions for processor **202**, for example.

Mass storage **207** can be used to store information and instructions. For example, hard disks such as the Adaptec® family of SCSI drives, an optical disc, an array of disks such as RAID (e.g. the Adaptec family of RAID drives), or any other mass storage devices may be used, for example. Bus **201** communicably couples processor(s) **202** with the other memory, storage and communication blocks. Bus **201** can be a PCI/PCI-X or SCSI based system bus depending on the storage devices used, for example. Removable storage media **205** can be any kind of external hard-drives, floppy drives, flash drives, IOMEGA® Zip Drives, Compact Disc-Read Only Memory (CD-ROM), Compact Disc-Re-Writable (CD-RW), or Digital Video Disk-Read Only Memory (DVD-ROM), for example. The components described above are meant to exemplify some types of possibilities. In no way should the aforementioned examples limit the scope of the invention, as they are only exemplary embodiments.

FIG. **3** depicts a flow diagram **300** illustrating an activation of response status through use of an easy response button, according to embodiments of the present invention. First, a user logs in to the system by providing information about the individual and/or entering authentication or credential information (block **302**). At the login screen, the user may input, select, or otherwise specify information such as specific company, username, and/or password, according to embodiments of the present invention. The user may be viewing any screen of the navigation device **110** (block **304**), and when an incident is assigned to the particular vehicle or crew with which navigation device **110** is associated (block **306**), the entire screen of the display device screen of navigation device **110** may be turned into an easy response button, which is illustrated in FIG. **5**. The easy response button screen of FIG. **5** may list the nature of the assigned incident (e.g. "Vehicle Fire") and/or the location of the incident (e.g. "Lyon Barbee B" or "Main Street Café") and/or a time that has elapsed since the incident was assigned to the particular crew.

As used herein, the word "screen" is used in some instances to refer to the physical display device capable of displaying images, for example the display device incorporated with navigation system **110** (e.g. "device screen"), and in other instances to refer to the image, for example the particular pattern of colors, buttons, text, arrangement, and/or other visual appearance, displayed on the device screen (e.g. "display screen"). For example, FIGS. **5** through **25** illustrate display screens, while device **110** of FIG. **1** shows a device screen.

The screen of navigation device **110** may be a touch screen device or a device with similar operational characteristics, to permit tactile gestures to be used to indicate selections on the screen. By touching or tapping anywhere on the screen of FIG. **5**, the user activates the easy response button (block **308**), after which the status of the particular vehicle or crew is set to "en route" or the like (block **310**), and the display screen is changed to the easy response screen illustrated in FIG. **12**. Although a particular workflow for an incident assignment is described, other workflows may be achieved with navigation device **110**.

FIG. **4** illustrates a navigational linking relationship map **400** between display screens of an EMS navigation device **110**, according to embodiments of the present invention. At login screen **402**, a user logs in to the system by providing information about the individual and/or entering authentication or credential information. At the login screen **402**, the user may input, select, or otherwise specify information such as specific company, username, and/or password, according to embodiments of the present invention. At the unit crew screen **404**, an example of which is illustrated at FIG. **22**, the user may specify the vehicle, unit, and/or additional crew members for the unit for which the navigation system **110** will receive incident assignments and messages, according to embodiments of the present invention.

At the first menu screen **406**, which is illustrated at FIG. **6**, a user may select additional screens and/or functionality to view. For example, a user may select the Dispatch button to view the dispatch screen(s) of FIGS. **7** to **11**, the Response button to view the easy response screen(s) of FIGS. **12** to **14**, the Work button to view the work screen(s) of FIGS. **15** and **16**, and the Chat Messaging button to view the chat messaging screen(s) of FIGS. **17** to **19**. Clicking on the Next: Menu (Page 2) button will take the user to the second menu screen of FIG. **20**.

Clicking on the Dispatch button from the home screen **406** takes the user to the first dispatch screen **408**, illustrated in FIG. **7**. The screen of FIG. **7** displays information about incident identifiers and type, and/or the location for the incident, for example the incident to which the ambulance is responding. From this screen **408**, clicking on the "Next Dispatch" button takes the user to the second dispatch screen **410**, illustrated at FIG. **8**, which also displays information about the incident identifiers and type, and/or the transport location for the incident. Clicking on the Next Dispatch button from screen **410** navigationally links the user to a third dispatch screen **412**, illustrated in FIG. **9**, which displays notes and alerts for the incident. The notes may include any entries added by dispatch throughout the lifecycle of the particular incident, according to embodiments of the present invention. Clicking on the Next Dispatch button from screen **412** navigationally links the user to a fourth dispatch screen **414**, illustrated in FIG. **10**, which displays patient information for the patient associated with the incident. This may include patient characteristics, referring doctor, prior incidents, home address, employer information, and/or payor information, according to embodiments of the present invention. Clicking on the Next Dispatch button from screen **414** takes the user to a fifth dispatch screen **416**, illustrated in FIG. **11**, which displays all time stamps entered for the incident, and/or duration on each unit status for the incident, according to embodiments of the present invention.

From the home menu **406**, selecting the Response button takes the user to easy response screen **418**, which is illustrated at FIG. **12**. Screen **418** displays a map with a route to the target destination, including address and directions. This screen **418** allows for setting time stamps for the incident, viewing alerts, viewing notes, and/or toggling the map view (for example between zooming to the destination location, viewing the entire route, and following the vehicle on the map), according to embodiments of the present invention. According to some embodiments of the present invention, the toggle button used to toggle between views is in the same location with respect to the display screen device in each of the two or three or more views, to facilitate easy and rapid toggling and/or cycling between the various kinds of views, without having to locate and select a different button for each view. Selecting the Next button from screen **418** takes the user to a second easy response screen **420**, as illustrated in FIG. **13**, which displays a second set of commands available on the easy response screen. These additional commands include switching to the dispatch screen **408** to view the incident details (Incident Details button), closing a road on the map, viewing a list of road closures, and/or routing to a custom address, according to embodiments of the present invention. Selecting the Next button from screen **420** takes the user to a third easy response screen **422**, illustrated at FIG. **14**, which displays a third set of commands including specifying which map layers to display on the map (Map Layers button), toggling traffic overlay on the map (Show Traffic button), and toggling weather overlay on the map (Show Weather button), according to embodiments of the present invention.

From the home menu **406**, selecting the Work button takes the user to work screen **424**, illustrated at FIG. **15**, which displays a set of incidents that have been assigned to the unit throughout the shift, including open and completed incidents, according to embodiments of the present invention.

Selecting a particular item from the incidents listed on screen **424** takes the user to work selection screen **426**, illustrated at FIG. **16**, which shows a summary of the selected incident in the content area, and allows the user to navigate to the dispatch screen **408** (View Full Details button) to view the full incident details, or to the easy response screen **418** (Route to Scene button) showing a route between the unit and the incident location, according to embodiments of the present invention.

From the home menu **406**, selecting the Chat Messaging button takes the user to the chat messaging screen **428**, illustrated at FIG. **17**, which displays a list of facility diversions and textual messages sent to and/or from other units and/or dispatch, according to embodiments of the present invention. From screen **428**, selecting a particular message from the list (e.g. by tapping or clicking on it) takes the user to a chat messaging details screen **430**, illustrated at FIG. **18**, which displays the details for a chat message including when the message was sent, the sender of the message, the full message text, and/or when the message expires (at which time it may be removed from the message list, for example). Selecting the Compose button from screen **428** takes the user to the chat messaging editor screen **432**, illustrated at FIG. **19**, which displays an editor interface for composing a chat message to send, according to embodiments of the present invention.

From the home menu **406**, selecting the Next:Menu button takes the user to the second menu screen **434**, which is illustrated at FIG. **20**. Pushing the Weather button from screen **434** takes the user to the weather screen **436**, illustrated at FIG. **21**, which displays the current weather conditions and/or projected weather conditions for a particular time period or throughout the day, according to embodiments of the present invention. Pushing the Unit Crew button from screen **434** takes the user to the unit crew screen **438** illustrated at FIG. **22**, which permits the user to change the vehicle, unit, and additional crew members for the user who is logged in to the system, according to embodiments of the present invention. Pushing the Logout button from screen **434** logs out the current user and takes the user to the login screen. Selecting the System Alerts button from screen **434** takes the user to the system alerts screen **442**, illustrated at FIG. **23**, which displays any system errors that have occurred, for example when the user-entered time stamp for an incident was rejected because it was already set by dispatch, according to embodiments of the present invention.

From the second menu screen **434**, selecting the Next: Menu button takes the user to the third menu screen **444**, as illustrated at FIG. **24**, according to embodiments of the present invention. Selecting the Next:Menu button from screen **444** takes the user back to the home screen **406**. Selecting the System Settings button from screen **444** takes the user to system settings screen **446**, illustrated at FIG. **25**, which allows the user to customize the features and behavior of the application. For example, when the "Easy Response on Assignment" checkbox is activated (e.g. by clicking on or tapping the checkbox), the user is taken immediately to the easy response screen **418** as soon as the easy response button (FIG. **5**) is activated. The easy response button screen of FIG. **5** may be color coded to draw attention to the fact that a new incident assignment has been received; for example, the easy response button screen of FIG. **5** may be colored bright red or orange.

The display screens described above show the primary navigation buttons as being of substantially the same size, and/or oriented adjacent to the right side of the screen display device. Activating the "Left Orientation" checkbox in screen **446** switches the orientation and alignment of the primary navigation buttons from the right side to the left side of the screen. This may be done for left-handed users, for example.

Placing the primary navigation buttons against either the right or left sides of the display screen facilitates user of the interface by EMS users. This feature permits an EMS user to place one or more fingers onto the physical device containing the display screen, and use his or her thumb to select the primary navigation buttons, which are rather large (and may be larger than any other buttons on the screens, for example secondary buttons which are not often needed during an incident response). For example, when the navigation buttons are placed adjacent to the right side perimeter of the display screen, a user may place a hand on the side of the device, with the fingers behind the device and/or pointed away from the user, while the user uses the thumb of the hand to navigate the menu structure described with respect to FIG. **4**, according to embodiments of the present invention. As such, the user can use the physical device as a reference point, without having to continually visually scan the display to find new button placements and/or to find smaller or more obscure buttons while navigating the screens or pages. The user knows that somewhere near the bottom right of the screen is the lowest button, and somewhere near the top right of the screen is the highest button. This same setup may be employed by placing the buttons along the left side of the screen. According to some embodiments, the buttons may be placed adjacent to either the top or the bottom edge of the screen, depending on the placement and orientation of the navigation device **110**, and the particular user's preferences.

Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. For example, while the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

What is claimed is:

1. A system for tracking and management of emergency medical services (EMS) responses, the system comprising one or more servers configured to:

during a shift of an EMS crew, assign at least an emergency non-prescheduled transport and a non-emergency pre-scheduled transport to one EMS vehicle of the EMS crew;

provide a user interface comprising a map comprising location information for the EMS vehicle;

receive location information updates for the EMS vehicle;

for each assigned transport, update the map and a status indication for the EMS vehicle at the user interface based on the location information updates,

wherein

the status indication update, for one location information update of the location information updates, comprises a transporting indication corresponding to a transport of a patient in the EMS vehicle, and

the map update comprises a transport destination for the patient in the EMS vehicle; and

store a status history for the EMS vehicle.

2. The system of claim **1**, wherein the one or more servers provide a client-server architecture.

3. The system of claim 1, comprising a mobile device associated with the EMS vehicle, wherein the mobile device is configured to display navigation information and incident information, wherein assigning each respective assigned transport comprises providing, for review at the mobile device, the incident information associated with the respective assigned transport.

4. The system of claim 3, wherein the mobile device provides user selectable controls corresponding to scene location information and destination location information.

5. The system of claim 3, wherein the mobile device is configured to provide, to at least one of the one or more servers, one or more of vehicle speed information, travel time, or an estimated time of arrival.

6. The system of claim 1, wherein the user interface comprises a chat messaging interface for chat messages between the EMS crew and a remote dispatch service.

7. The system of claim 6, wherein the chat messaging interface comprises a list of textual messages.

8. The system of claim 6, wherein the chat messaging interface displays at least one of a message time stamp, a sender identification, or a full text associated with a particular message.

9. The system of claim 6, wherein the chat messaging interface comprises a message editing screen configured to enable a user to compose a chat message.

10. The system of claim 1, wherein the location information comprises a name of an incident location.

11. The system of claim 1, wherein the location information comprises global positioning system (GPS) information.

12. The system of claim 1, wherein the EMS vehicle comprises an ambulance.

13. The system of claim 1, wherein the location information updates are real-time location information updates.

14. The system of claim 1, wherein the user interface comprises a dispatch screen.

15. The system of claim 1, wherein the status indication update for another information update of the location information updates comprises one of an assigned indication, an open indication, an en route indication, an at scene indication, a transporting indication, an at destination indication, a canceled indication, or a complete indication.

16. The system of claim 1, wherein the one or more servers are configured to populate the map with a visual representation of an entire route for the EMS vehicle, the entire route comprising an incident location and a destination for patient transport.

17. The system of claim 16, wherein the entire route includes routing to a custom address.

18. The system of claim 1, wherein the user interface is configured to provide one or more of incident details, dispatch notes, road closure information, or weather information.

19. The system of claim 18, wherein the dispatch notes comprise one or more of patient characteristics, physician information, prior incident information for the patient in the EMS vehicle, a patient home address, a patient employer, or payor information for the patient in the EMS vehicle.

20. The system of claim 1, wherein the one or more servers are configured to associate run numbers and determinant codes with each of respective assigned transport.

21. The system of claim 1, wherein the status indication comprises one or more time stamps associated with each respective assigned transport, wherein

the one or more time stamps indicate one or more response times for a crew assigned to the respective assigned transport.

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